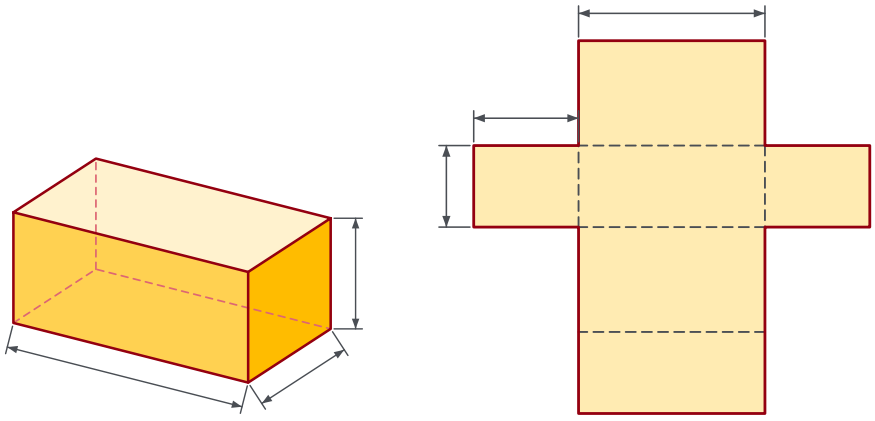
ACCESS TRADES

Trades Numeracy Exercise Book

Practice for every chapter of the Trades Numeracy Textbook: skills, shop problems and Shape Canvas drawings, in metric and imperial.



ACCESS

Skills for Success: Numeracy
Answers are in the Answer Key

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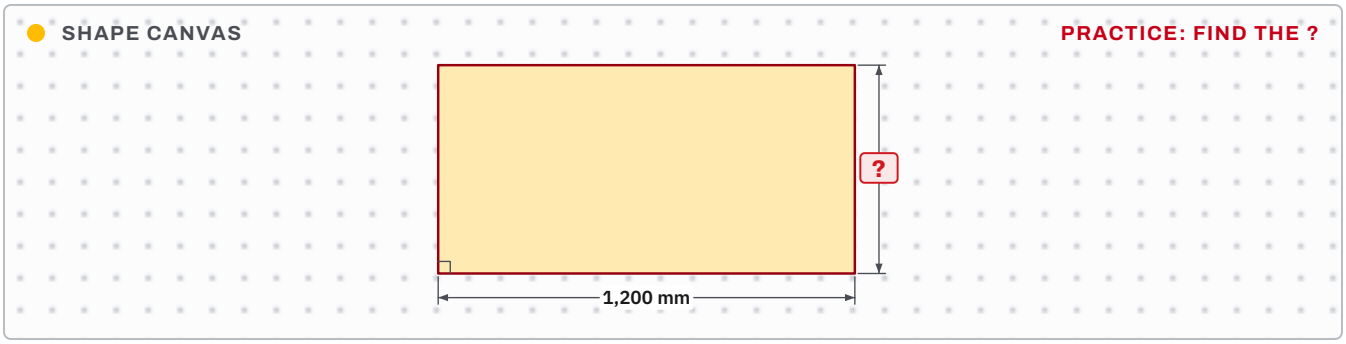
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How to use this book

Each chapter of this book matches the chapter with the same number in the Trades Numeracy Textbook. Work through the textbook chapter first, then use these exercises for more practice.

Every chapter has three sets. Set A practises the skills one at a time. Set B puts them to work in shop problems. Set C uses Shape Canvas drawings: a red ? marks a hidden size to find, dashed lines break a part into simple shapes, and a solid shown beside its flat pattern is unfolded into its net.

Show your work in the working box under each problem and write the answer on the answer line. Estimate first, then use your calculator. The keys for the TI-30Xa and the Sharp EL-531XT are shown in the textbook. Answers to every problem are in the Answer Key.



1 Whole Numbers

EXERCISES

Practice with place value, rounding and estimating, the four operations, averages and multi-step shop problems. Estimate each answer first, then check it with your calculator.

EXERCISE 1A

Skills

Place value, rounding, estimating and the four operations

Name _____

Date _____

1 Write 4,072 in words.

WORKING

Answer _____

2 Write in digits: fifteen thousand three hundred eight.

WORKING

Answer _____

3 What is the 7 worth in 37,215?

WORKING

Answer _____

4 Put in order, smallest first: 3,409, 3,490, 3,049, 3,940.

WORKING

Answer _____

5 Round 3,462 to the nearest ten.

WORKING

Answer _____

6 Round 8,951 to the nearest hundred.

WORKING

Answer _____

7 Round 14,500 to the nearest thousand.

WORKING

Answer _____

8 Round 795 to the nearest ten.

WORKING

Answer _____

9 Estimate 1,180 + 2,940 + 615 by rounding each number to the nearest hundred.

WORKING

Answer _____

10 Estimate 48 × 21 by rounding each number to the nearest ten.

WORKING

Answer _____

11 3,275 + 1,948

WORKING

Answer _____

12 509 + 2,716 + 88

WORKING

Answer _____

13 7,000 – 2,368

WORKING

Answer _____

14 10,250 – 4,875

WORKING

Answer _____

15 32 × 15

WORKING

Answer _____

16 215 × 24

WORKING

Answer _____

17 2,688 ÷ 12

WORKING

Answer _____

18 Divide 5,000 by 320. Give the whole-number quotient and the remainder.

WORKING

Answer _____

19 Find the average of 64, 58, 71 and 67.

WORKING

Answer _____

20 Check 7,000 – 2,368 by adding. What should you get?

WORKING

Answer _____

EXERCISE 1B

Shop problems

Whole numbers on the job, in metric and imperial

Name _____

Date _____

1 Three pieces, 1,350 mm, 980 mm, 2,145 mm, are cut from a 6,000 mm length of flat bar. Ignore the saw cuts. How long is the offcut?

WORKING

Answer _____

2 A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

WORKING

Answer _____

3 A millwright sets 14 baseplates. Each baseplate takes 8 anchor bolts. Anchor bolts come in boxes of 25. How many bolts are needed, and how many boxes?

WORKING

Answer _____

4 A pipe job has 9 flanged joints. Each joint takes 8 bolts and each bolt takes 2 nuts. How many nuts are needed?

WORKING

Answer _____

5 Lengths of 18 in are cut from a 20 ft bar of angle (20 ft = 240 in). Ignore the saw cuts. How many pieces can be cut, and how long is the offcut?

WORKING

Answer _____

6 A 48 in bar has 7 holes in a straight row. The end holes are 3 in from the ends and the holes are evenly spaced. Find the spacing, centre to centre.

WORKING

Answer _____

7 A welder logs 7, 9, 8, 6, 10 hours over five days. What is the average number of hours per day?

WORKING

Answer _____

8 A machine makes 45 parts an hour. It runs 3 shifts of 8 hours. How many parts does it make? Estimate first, using 50 parts an hour.

WORKING

Answer _____

9 A crew uses 18 welding rods a day for 12 days. Rods come in boxes of 50. How many boxes do they need, and how many rods are left over?

WORKING

Answer _____

EXERCISE 1C

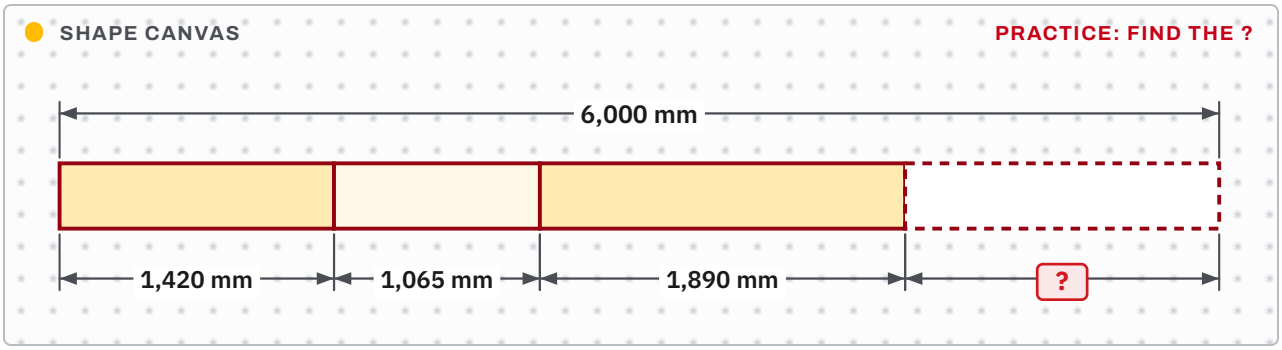
Shape Canvas

Find each red ? on the drawing

Name _____

Date _____

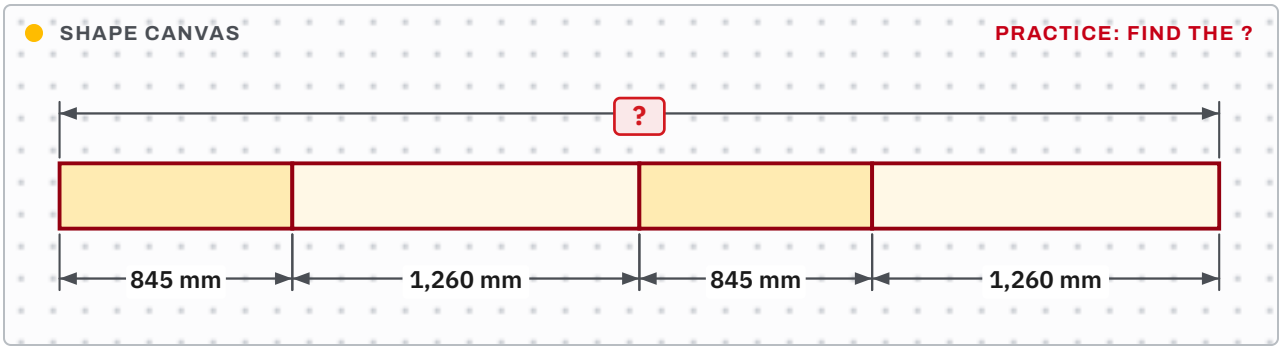
1 Three pieces are cut from a 6,000 mm bar. Ignore the saw cuts. Find the offcut.



WORKING

Answer _____

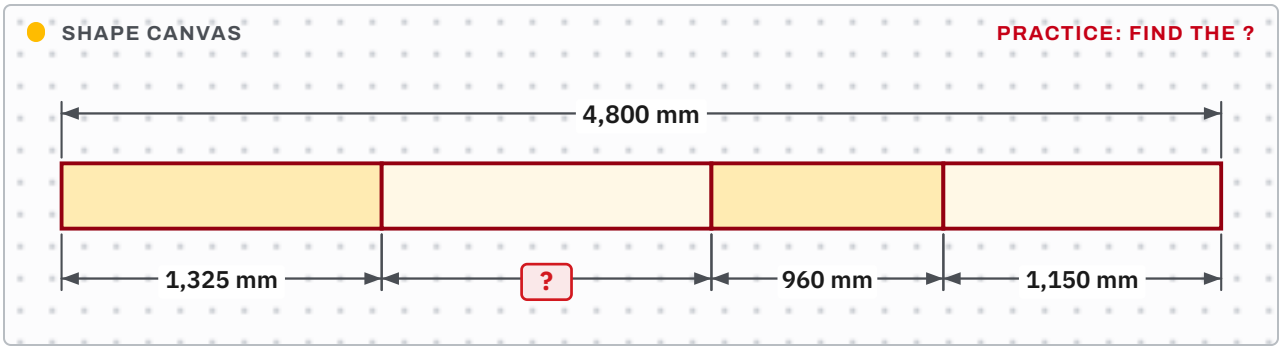
2 Find the overall length.



WORKING

Answer _____

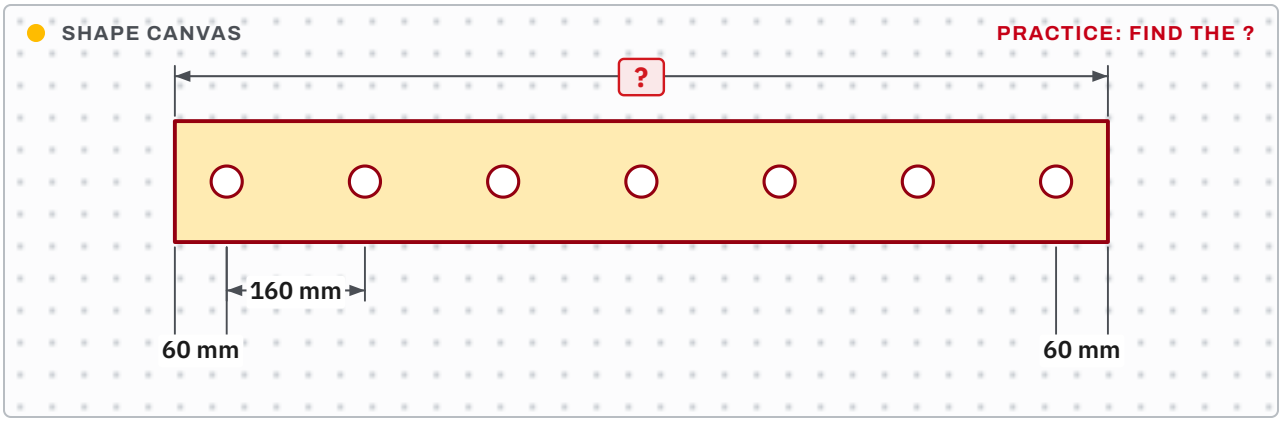
3 The overall length is 4,800 mm. Find the hidden piece.



WORKING

Answer _____

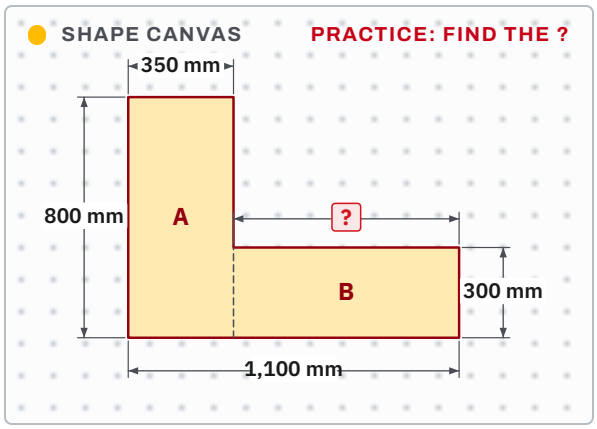
4 Seven holes are evenly spaced 160 mm apart along a plate. The end holes are 60 mm from the ends. Find the length of the plate.



WORKING

Answer _____

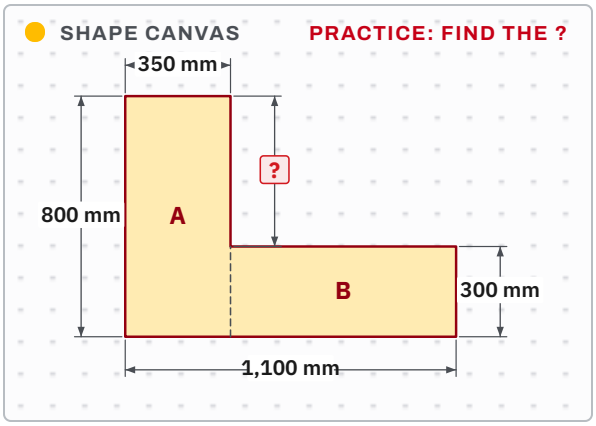
5 Find the hidden width. The dashed line splits the plate into A and B.



WORKING

Answer _____

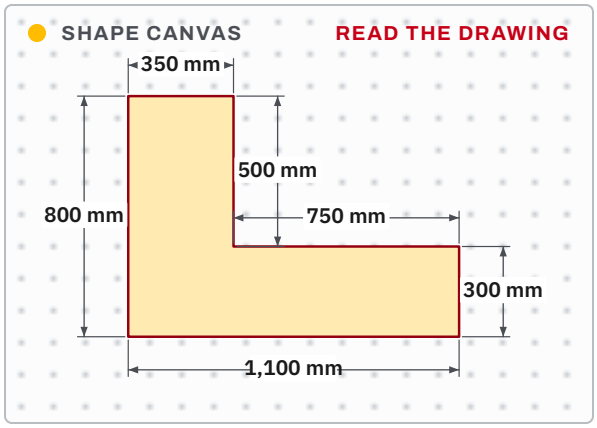
6 Find the hidden height of the upright part A.



WORKING

Answer _____

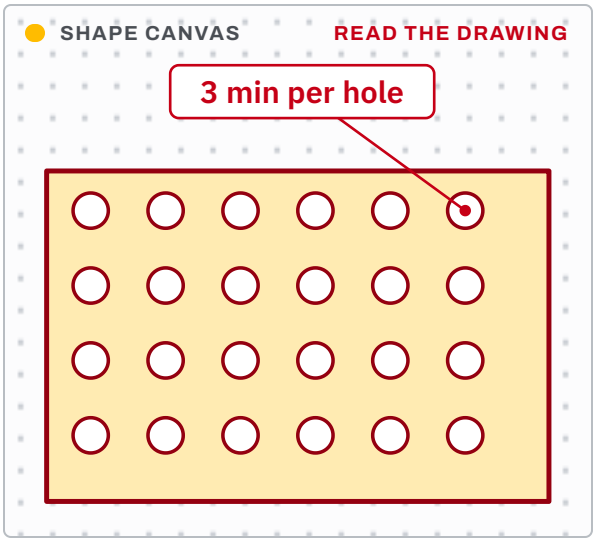
7 Find the distance around the plate: add all six edges.



WORKING

Answer _____

8 Count the holes with one multiplication. How long does the plate take?



WORKING

Answer _____

9 Change one value: each of the three pieces in problem 1 is cut 25 mm longer. How much shorter is the offcut now?

WORKING

Answer _____

2 Your Calculator: TI-30Xa and Sharp EL-531XT

EXERCISES

Practice with the keys of both calculators: writing key sequences, second functions, negative numbers, powers and roots, π, fractions, percent, memory and the trig keys. Start every problem with ON/C ON/C and DEG on the display.

EXERCISE 2A

Skills

Write the keys for both calculators, then the answer

Name _____

Date _____

1 3,640 + 1,295 − 788

WORKING

Answer _____

2 23^2

WORKING

Answer _____

3 $\sqrt{1{,}369}$

WORKING

Answer _____

4 7^3

WORKING

Answer _____

5 The cube root of 343

WORKING

Answer _____

6 (−15) × 6 + 40

WORKING

Answer _____

7 3^5, using the y^x key

WORKING

Answer _____

8 π × 90, to two decimal places

WORKING

Answer _____

9 $2\frac{3}{8} + 1\frac{7}{16}$, using the fraction key. Write the answer as the display shows it.

WORKING

Answer _____

10 Change $\frac{7}{8}$ to a decimal.

WORKING

Answer _____

11 18% of 450

WORKING

Answer _____

12 1,250 plus 8%

WORKING

Answer _____

13 sin 52°, to four decimal places

WORKING

Answer _____

14 The angle whose tangent is 0.6, to two decimal places

WORKING

Answer _____

15 Store 640 + 385 in memory, clear the display, then find 2,400 minus the stored number.

WORKING

Answer _____

16 1 ÷ 8, using the 1/x key (x^{-1} on the Sharp)

WORKING

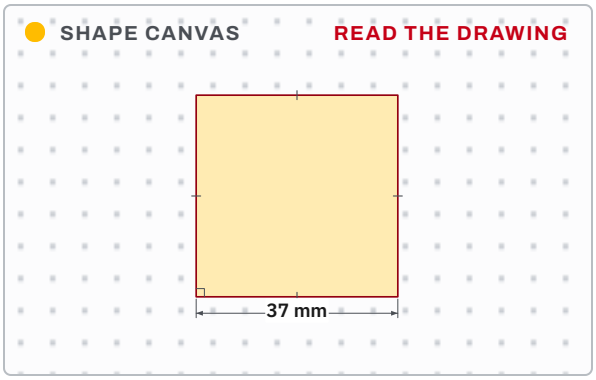
Answer _____

17 A learner needs (−25) + 9 and presses the minus key to make the 25 negative. Which key should they use instead on each calculator? What is the answer?

WORKING

Answer _____

18 Find the area of the square with the x^2 key.



WORKING

Answer _____

EXERCISE 2B

Shop problems

Use the calculator keys on the job

Name _____

Date _____

1 A pipe has an outside diameter of 114.3 mm. The distance around it is π × D. Find it to one decimal place.

WORKING

Answer _____

2 The round lid of a tank has a radius of 600 mm. Its area is π × r². Find it to the nearest square millimetre.

WORKING

Answer _____

3 Three pieces, each $14\frac{5}{8}$ in long, are cut from a 72 in bar. Ignore the saw cuts. Use the fraction key to find the offcut: $72 - 3 \times 14\frac{5}{8}$.

WORKING

Answer _____

4 A job needs 380 m of cable. The order adds 12% for waste. Use the % key to find the length to order.

WORKING

Answer _____

5 Each frame has four welds: 1,240 mm, 860 mm, 1,240 mm and 860 mm. Store the weld length for one frame in memory, then use it to find the weld length for 18 frames.

WORKING

Answer _____

6 The diagonal of a 1,200 mm by 900 mm frame is $\sqrt{(1{,}200^2 + 900^2)}$. Find it. Write the keys for both calculators.

WORKING

Answer _____

7 A ramp rises 450 mm over a run of 2,400 mm. Its angle is the angle whose tangent is 450 ÷ 2,400. Find it to one decimal place. Check that DEG is on the display first.

WORKING

Answer _____

8 A gauge reads −11. The setting it should read is 4. Find −11 − 4 to see how far off it is. Use the change-sign key.

WORKING

Answer _____

EXERCISE 2C

Shape Canvas

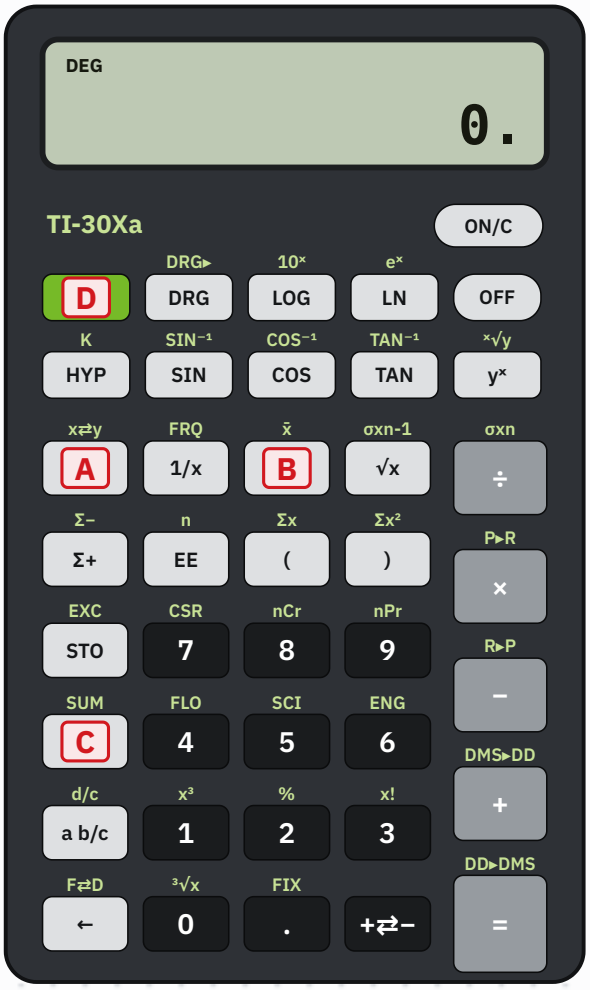
Keys, displays and drawings

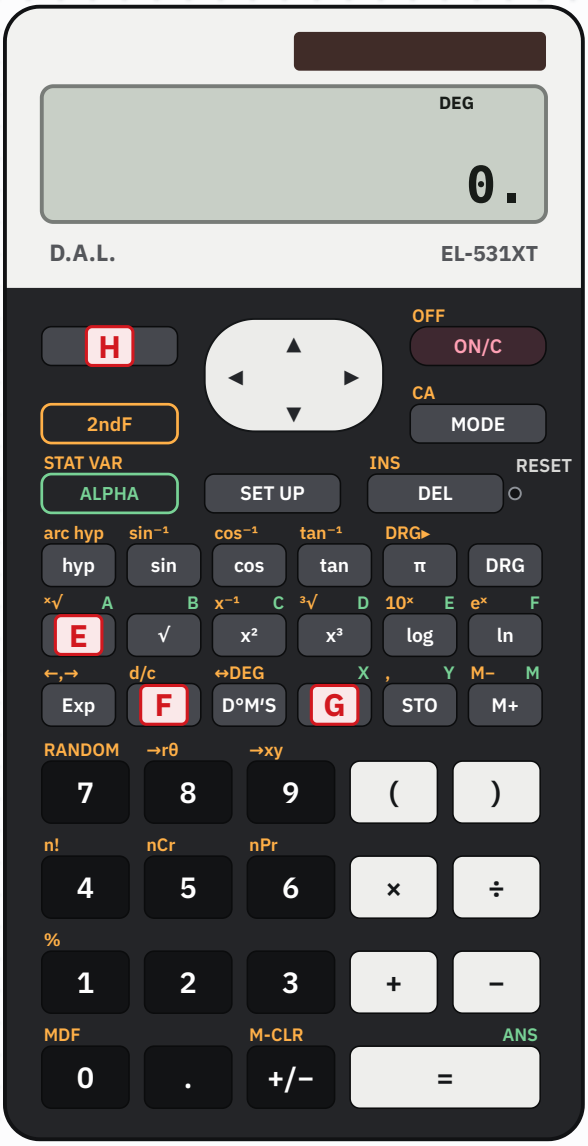
Name _____

Date _____

1 Name keys A to H and say what each one does. A to D are on the TI-30Xa, E to H on the Sharp EL-531XT.

SHAPE CANVAS — NAME THE KEYS



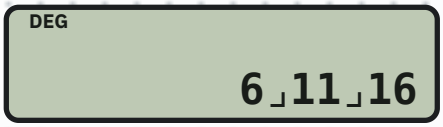


WORKING

Answer _____

2 Write the number this display shows.

SHAPE CANVAS READ IT



TI-30Xa

WORKING Answer _____

3 Write the number this display shows, with a comma.

SHAPE CANVAS READ IT



Sharp EL-531XT

WORKING Answer _____

4 What calculation was keyed? Write it with an exponent.

SHAPE CANVAS READ IT



Sharp EL-531XT

WORKING Answer _____

5 The small number at the right is an exponent of 10. Write the number in full, with commas.

SHAPE CANVAS READ IT



TI-30Xa

WORKING Answer _____

6 This display shows GRAD. How many presses of DRG bring back DEG?

SHAPE CANVAS READ IT

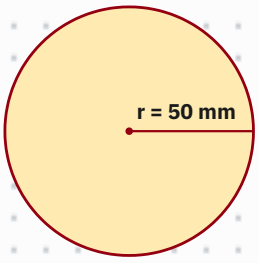


Sharp EL-531XT

WORKING Answer _____

7 Find the distance around the circle, π × D, to two decimal places. Then change one value: if the diameter doubles, what is the new distance around?

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

3 Order of Operations

EXERCISES

Practice with BEDMAS: brackets, exponents and roots, × and ÷ left to right, + and − left to right, and the fraction bar. Work each problem on paper first, then key it so the calculator follows the same order.

EXERCISE 3A

Skills

Work each one out with BEDMAS

Name _____

Date _____

1 $9 + 6 \times 4$

WORKING

Answer _____

2 $50 - 18 \div 3$

WORKING

Answer _____

3 $7 \times 8 - 6 \times 5$

WORKING

Answer _____

4 $(15 - 7) \times 4$

WORKING

Answer _____

5 $36 \div (4 + 5)$

WORKING

Answer _____

6 $[30 - (8 + 7)] \div 5$

WORKING

Answer _____

7 $2 \times [12 + (9 - 3)]$

WORKING

Answer _____

8 $6^2 - 4 \times 5$

WORKING

Answer _____

9 $(6 - 4)^2 \times 5$

WORKING

Answer _____

10 3×4^2

WORKING

Answer _____

11 $\sqrt{(144 - 44)}$

WORKING

Answer _____

12 $\sqrt{49} + \sqrt{81}$

WORKING

Answer _____

13 $72 \div 6 \div 3$

WORKING

Answer _____

14 $15 \div 3 \times 5$

WORKING

Answer _____

15 $40 - 15 + 5$

WORKING

Answer _____

16 $2 + 3^3$

WORKING

Answer _____

17 $\frac{56 + 28}{9 - 2}$

WORKING

Answer _____

18 $\frac{3 \times 15 + 5}{10 \times 2}$

WORKING

Answer _____

19 Add one pair of brackets to make this true: $4 + 6 \times 3 = 30$.

WORKING

Answer _____

20 Add one pair of brackets to make this true: $24 \div 6 - 2 = 6$.

WORKING

Answer _____

EXERCISE 3B

Shop problems

Write one calculation for each job, then work it out

Name _____

Date _____

1 A spool piece has 6 flanges with 8 bolts each. You also want 4 spare bolts. Write one calculation for the bolts needed and work it out.

WORKING

Answer _____

2 A rectangular guard frame is 1,350 mm by 875 mm. The distance around it is $2 \times (1{,}350 + 875)$. Find it.

WORKING

Answer _____

3 Four 27 in pieces and two 15 in pieces are cut from a 144 in bar. Ignore the saw cuts. The offcut is $144 - (4 \times 27 + 2 \times 15)$. Find it.

WORKING

Answer _____

4 Three pipe spools measure 3,250 mm, 2,980 mm and 3,105 mm. Their average length is $\frac{3{,}250 + 2{,}980 + 3{,}105}{3}$. Find it to two decimal places.

WORKING

Answer _____

5 A 60 in bar has 5 evenly spaced holes, with the end holes 4 in from the ends. The spacing is $\frac{60 - 2 \times 4}{4}$. Find it.

WORKING

Answer _____

6 Three pipe pieces of 1,200 mm, 850 mm and 640 mm go into a run with 6 fittings. For this job each fitting takes up 25 mm of the run. The length of the run is (1,200 + 850 + 640) + 6 × 25. Find it.

WORKING

Answer _____

7 A crew of 3 works 8 hours and a crew of 2 works 10 hours. The total hours are 3 × 8 + 2 × 10. A learner keys (3 × 8 + 2) × 10 instead. Find both answers and say which is right.

WORKING

Answer _____

8 A learner finds a hole spacing by keying 1,500 − 300 ÷ 4 = and gets 1,425. The spacing should be $\frac{1{,}500 - 300}{4}$. What went wrong? What are the right keys and the right answer?

WORKING

Answer _____

EXERCISE 3C

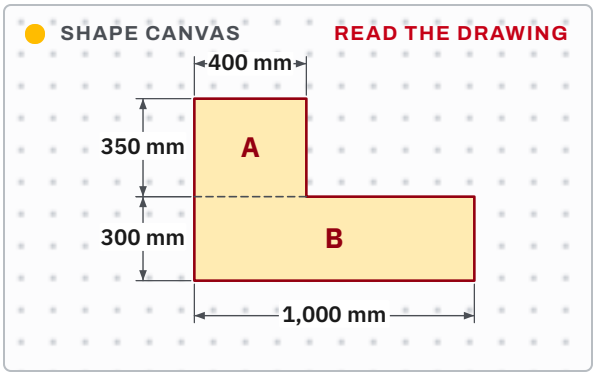
Shape Canvas

Write the calculation for each drawing, then work it out

Name _____

Date _____

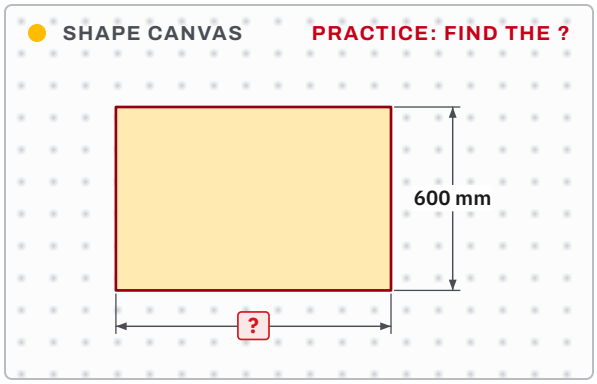
1 The plate is split into A and B. Its area in mm² is 400 × 350 + 1,000 × 300. Work it out.



WORKING

Answer _____

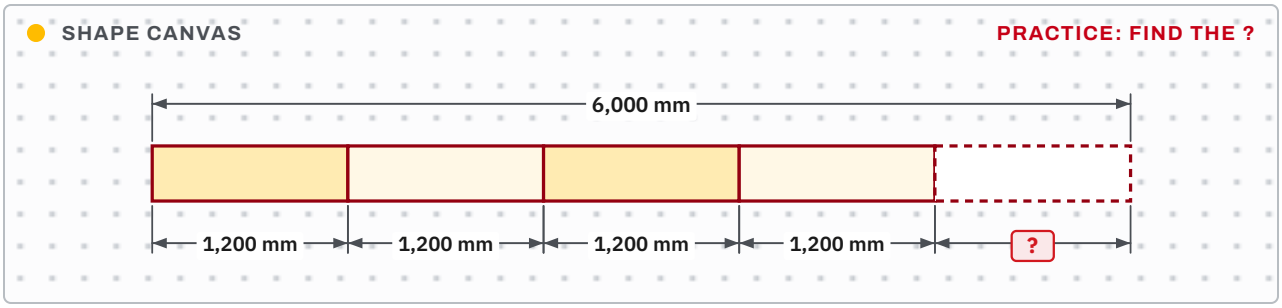
2 The distance around this frame is 3,000 mm. Find the hidden length with L = 3,000 ÷ 2 − 600.



WORKING

Answer _____

3 Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.

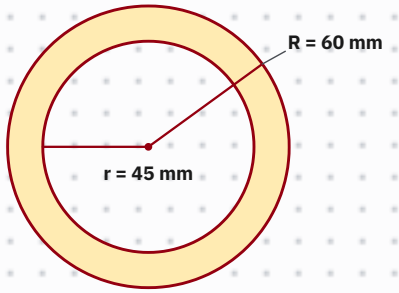


WORKING

Answer _____

4 The ring has R = 60 mm and r = 45 mm. Its area is π × (R² − r²). Find it to two decimal places. Then change one value: what is the area if r becomes 30 mm?

SHAPE CANVAS — READ THE DRAWING

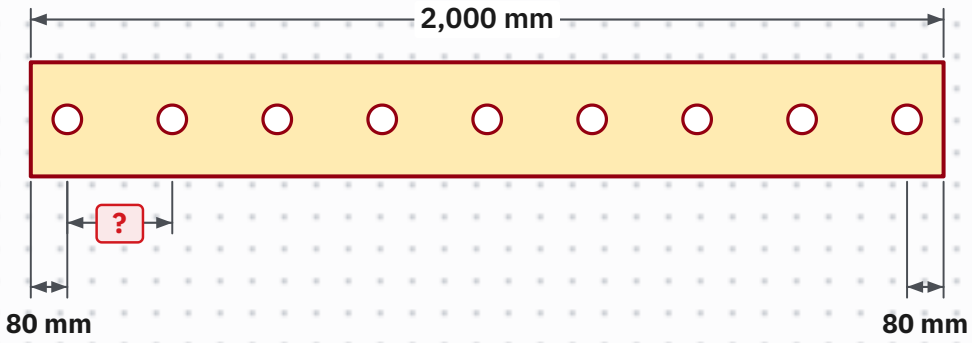


WORKING

Answer _____

5 Nine holes are evenly spaced on a 2,000 mm plate. The spacing is (2,000 − 2 × 80) ÷ 8. Find it.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

4 Fractions

EXERCISES

Practice reading tapes and rules, writing fractions in lowest terms, changing between mixed numbers and improper fractions, and adding, subtracting, multiplying and dividing fractions. Give every answer in lowest terms.

EXERCISE 4A

Skills

One skill at a time, from reading a rule to dividing fractions

Name _____

Date _____

1 What fraction of the bar is shaded? Is it a proper fraction?

SHAPE CANVAS READ THE DRAWING

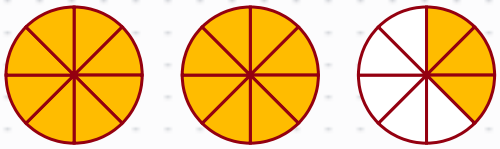


WORKING

Answer _____

2 What fraction of the circles is shaded? Write it as an improper fraction and as a mixed number.

SHAPE CANVAS READ THE DRAWING

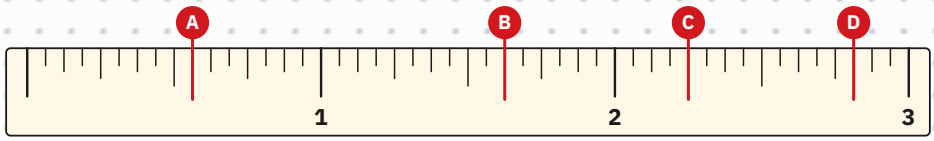


WORKING

Answer _____

3 Read marks A, B, C and D to the nearest $\frac{1}{16}$ in.

SHAPE CANVAS READ THE DRAWING

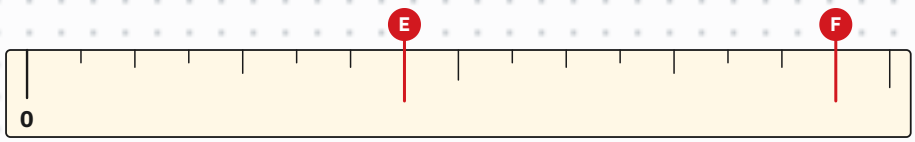


WORKING

Answer _____

4 This half inch of a rule is marked in 32nds. Read marks E and F.

SHAPE CANVAS READ THE DRAWING

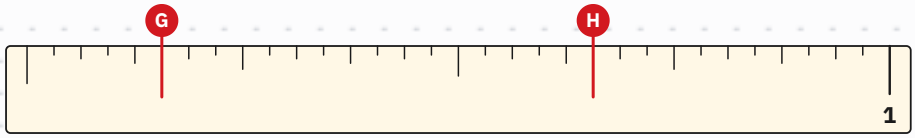


WORKING

Answer _____

5 This part of a rule runs from $\frac{1}{2}$ in to 1 in and is marked in 64ths. Read marks G and H.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

6 Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

WORKING

Answer _____

7 Write $\frac{7}{8}$ in 32nds and $\frac{5}{16}$ in 64ths.

WORKING

Answer _____

8 Which is longer, $\frac{9}{16}$ in or $\frac{5}{8}$ in? By how much?

WORKING

Answer _____

9 Put in order from smallest: $\frac{13}{32}$, $\frac{3}{8}$, $\frac{7}{16}$.

WORKING

Answer _____

10 Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

WORKING

Answer _____

11 Change $4\frac{7}{16}$ and $6\frac{3}{4}$ to improper fractions.

WORKING

Answer _____

12 The Sharp shows 3 ┌9 ┌16. Write the number as a mixed number. What does the TI-30Xa show after 2nd d/c?

WORKING

Answer _____

13 What is the reciprocal of $2\frac{1}{2}$?

WORKING

Answer _____

14 Add. (a) $\frac{5}{16} + \frac{3}{4}$ (b) $7\frac{3}{8} + 2\frac{15}{16}$

WORKING

Answer _____

15 Subtract. (a) $5 - 1\frac{11}{16}$ (b) $9\frac{1}{8} - 4\frac{3}{4}$

WORKING

Answer _____

16 Multiply. (a) $\frac{5}{8} \times 12$ (b) $2\frac{1}{2} \times 1\frac{3}{4}$

WORKING

Answer _____

17 Divide. (a) $\frac{3}{4} \div \frac{3}{16}$ (b) $15\frac{3}{4} \div 3$

WORKING

Answer _____

EXERCISE 4B

Shop problems

Fractions in fabrication, pipe trades and millwright work

Name _____

Date _____

1 A fabricator cuts pieces of $27\frac{5}{8}$ in, $19\frac{3}{16}$ in, $33\frac{1}{2}$ in from a 96 in bar. There are 3 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

2 A pipe fitter needs two pieces of pipe, $18\frac{3}{4}$ in and $22\frac{7}{16}$ in long. What total length of pipe is that, ignoring the saw cuts?

WORKING

Answer _____

3 A bolt passes through a $\frac{5}{8}$ in plate, a $\frac{3}{4}$ in plate and a $\frac{1}{8}$ in washer. What is the total thickness it clamps?

WORKING

Answer _____

4 A millwright needs to fill a $\frac{3}{16}$ in gap with $\frac{1}{32}$ in shims. How many shims does it take?

WORKING

Answer _____

5 A plate is $15\frac{5}{8}$ in wide. How far from the edge is the centre line?

WORKING

Answer _____

6 Nine holes are evenly spaced along a 48 in angle. The end holes are $1\frac{3}{4}$ in from the ends. Find the spacing, centre to centre.

WORKING

Answer _____

7 How many $4\frac{3}{8}$ in spacers can be cut from a 70 in bar? You make one cut for each spacer and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

8 A pipe has an outside diameter of $2\frac{3}{8}$ in and a wall $\frac{3}{16}$ in thick. What is the inside diameter?

WORKING

Answer _____

9 A keyway measures $1\frac{11}{16}$ in wide on the rule. Will a $1\frac{3}{4}$ in key fit? By how much is it too wide or too narrow?

WORKING

Answer _____

10 A 2,400 mm length of channel is cut so one piece is $\frac{3}{4}$ of the length. How long is that piece?

WORKING

Answer _____

11 A 45 in bar is cut so one piece is $\frac{2}{3}$ of the bar. How long are the two pieces? Ignore the saw cut.

WORKING

Answer _____

12 A 50 in bar is cut into 3 equal pieces. Ignore the saw cuts. How long is each piece, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

EXERCISE 4C

Shape Canvas

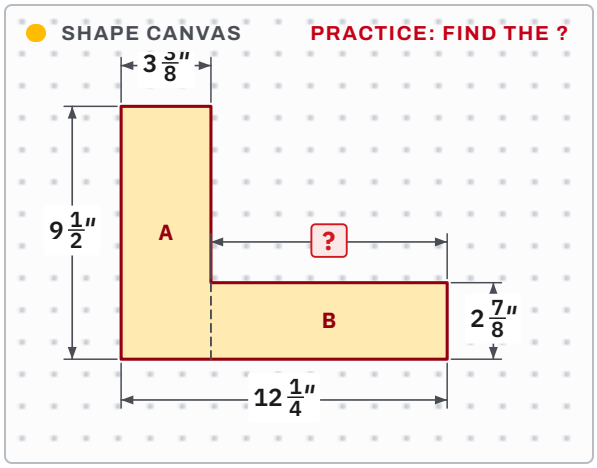
Parts in inches. Find every red ?

Name _____

Date _____

Each drawing is to scale. Break the part into pieces you know, and show the fractions you add or take away.

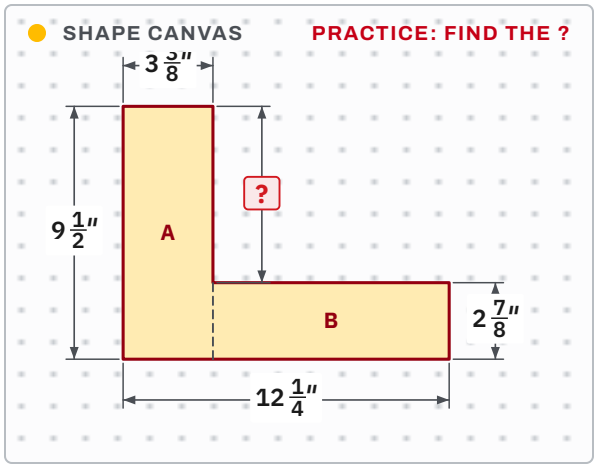
1 Find the hidden width.



WORKING

Answer _____

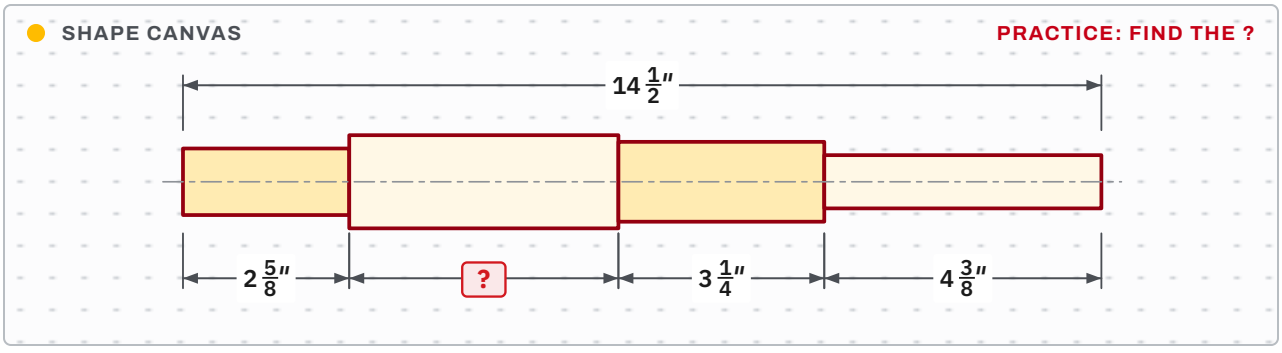
2 Find the hidden height.



WORKING

Answer _____

3 The shaft is $14\frac{1}{2}$ in long overall. Find the missing step.

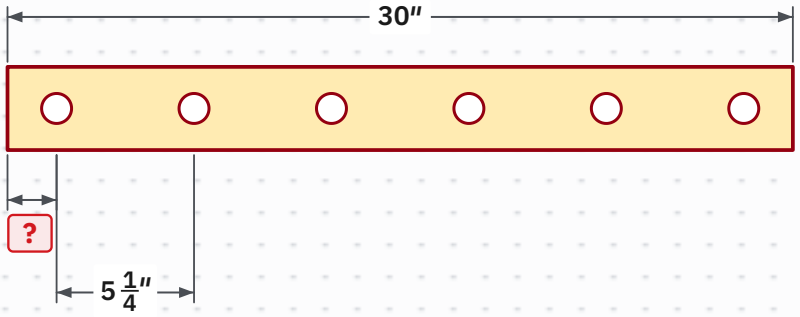


WORKING

Answer _____

4 Six holes are evenly spaced along a 30 in plate. Find the distance from each end to the first hole.

SHAPE CANVAS — PRACTICE: FIND THE ?

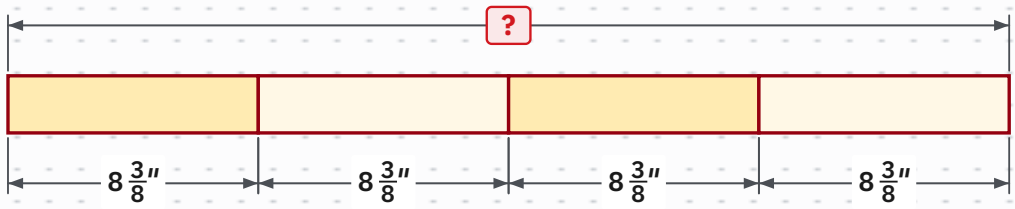


WORKING

Answer _____

5 Change one value: four equal pieces of $8\frac{3}{8}$ in make up the bar. Find the overall length. If each piece is made $\frac{1}{16}$ in longer, what is the new overall length?

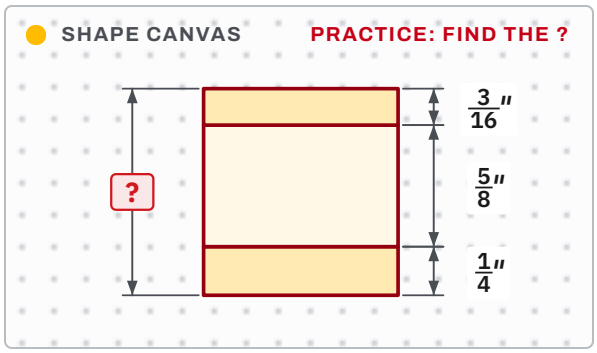
SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

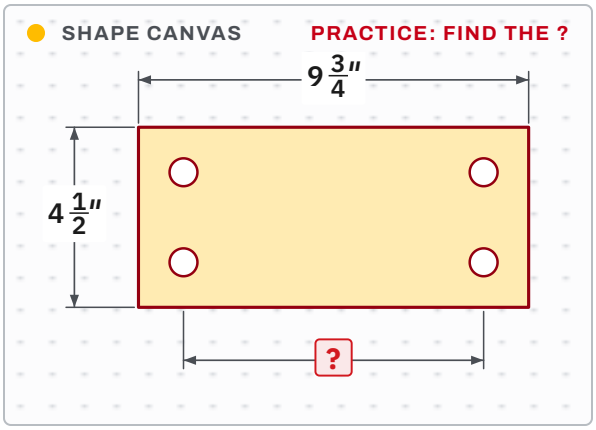
6 Three plates are stacked. How thick is the stack?



WORKING

Answer _____

7 Each hole centre is $1\frac{1}{8}$ in in from two edges. Find the centre-to-centre distance along the plate and across it.



WORKING

Answer _____

5 Decimals

EXERCISES

Practice place value, reading a metric rule, rounding, the four operations with decimals, changing between fractions and decimals, and money. Round only where the problem says so.

EXERCISE 5A

Skills

Place value, rounding, the four operations and conversions

Name _____

Date _____

1 What is the 4 worth in 30.249? Write 0.075 in words.

WORKING

Answer _____

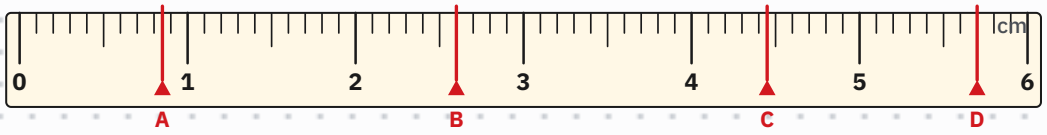
2 Write $\frac{3}{100}$ and $\frac{48}{1{,}000}$ as decimals.

WORKING

Answer _____

3 Read marks A, B, C and D to the nearest 0.5 mm.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

4 Put in order from smallest: 0.4, 0.375, 0.41.

WORKING

Answer _____

5 Which is longer, 2.5 mm or 2.45 mm? By how much?

WORKING

Answer _____

6 Round 7.846 to one decimal place and to two decimal places.

WORKING

Answer _____

7 Round 31.27 mm and 18.8 mm to the nearest 0.5 mm.

WORKING

Answer _____

8 Add. 63.5 + 9.53 + 0.8

WORKING

Answer _____

9 Subtract. 50 − 17.46

WORKING

Answer _____

10 Multiply. (a) 0.6 × 0.4 (b) 25.4 × 7 (c) 3.125 × 1,000

WORKING

Answer _____

11 Divide. (a) 8.4 ÷ 0.12 (b) 45.72 ÷ 6 (c) 19.5 ÷ 100

WORKING

Answer _____

12 Change $\frac{7}{8}$ and $3\frac{1}{4}$ to decimals.

WORKING

Answer _____

13 Change 0.875 and 0.08 to fractions in lowest terms.

WORKING

Answer _____

14 Change 1.47 in to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{32}$ in.

WORKING

Answer _____

15 Change 0.29 in to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

16 Change 30 mm to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

17 Change $1\frac{1}{4}$ in to millimetres.

WORKING

Answer _____

EXERCISE 5B

Shop problems

Decimals in fabrication, pipe trades and millwright work

Name _____

Date _____

1 Pieces of 412.5 mm, 287.25 mm, 650 mm are cut from a 1,500 mm bar. There are 3 cuts and the kerf is 2.5 mm. How long is the offcut?

WORKING

Answer _____

2 A pipe fitter needs 3 lengths of pipe, each 1.85 m. The pipe costs $14.60 per metre. What does the pipe cost?

WORKING

Answer _____

3 A millwright stacks shims that are each 0.25 mm thick. How many shims make a stack 1.75 mm thick?

WORKING

Answer _____

4 An order has 25 bolts at $0.46 each and 25 nuts at $0.12 each. What is the total?

WORKING

Answer _____

5 Flat bar on a job weighs 2.36 kg per metre. What do 4.5 m weigh?

WORKING

Answer _____

6 Eight holes are evenly spaced along a 900 mm plate. The end holes are 37.5 mm from the ends. Find the spacing to two decimal places.

WORKING

Answer _____

7 A drawing gives a length of 3.38 in. What do you mark on the tape, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

8 A metric part is 120 mm long. What is that to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

9 A pre-apprentice works 37.5 hours at $21.40 an hour. What is the pay before deductions?

WORKING

Answer _____

10 A $\frac{5}{8}$ in bolt hole is drilled in a metric part. What is the hole size in millimetres?

WORKING

Answer _____

EXERCISE 5C

Shape Canvas

Decimal sizes on the canvas. Find every red ?

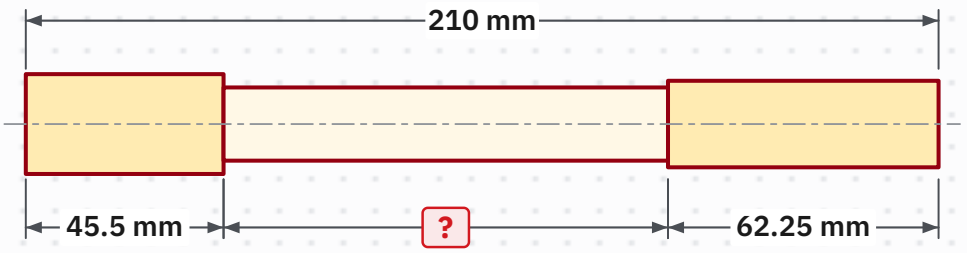
Name _____

Date _____

Each drawing is to scale. Line up the decimal points when you add or subtract.

1 The pin is 210 mm long overall. Find the length of the middle step.

SHAPE CANVAS — PRACTICE: FIND THE ?

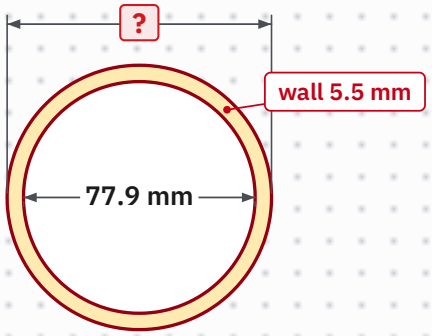


WORKING

Answer _____

2 Find the outside diameter of the pipe.

SHAPE CANVAS — PRACTICE: FIND THE ?

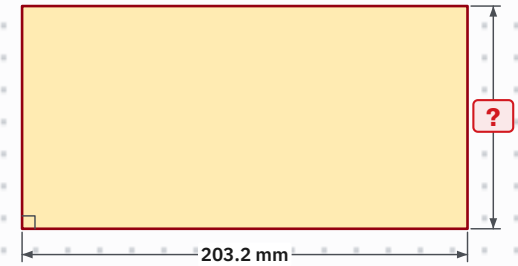


WORKING

Answer _____

3 The distance around the plate is 609.6 mm. Find the width.

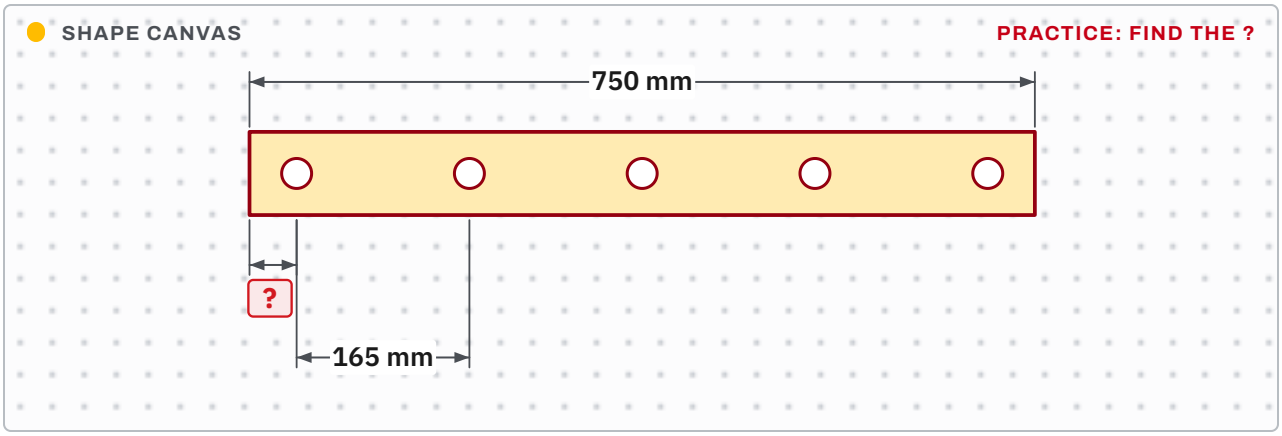
SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

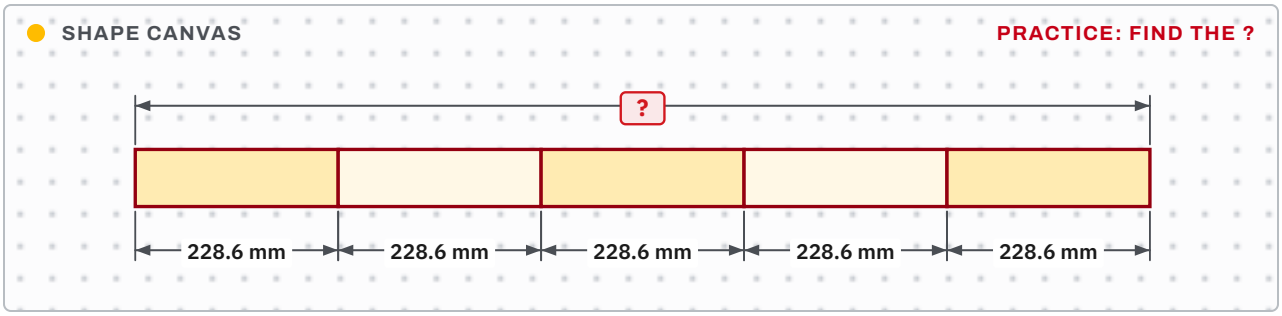
4 Five holes are evenly spaced along a 750 mm bar. Find the distance from each end to the first hole.



WORKING

Answer _____

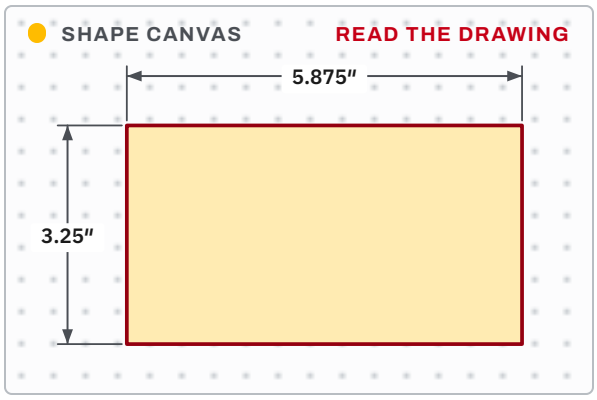
5 Change one value: five pieces of 228.6 mm make up the bar. Find the overall length. If each piece is cut 1.5 mm shorter, what is the new overall length?



WORKING

Answer _____

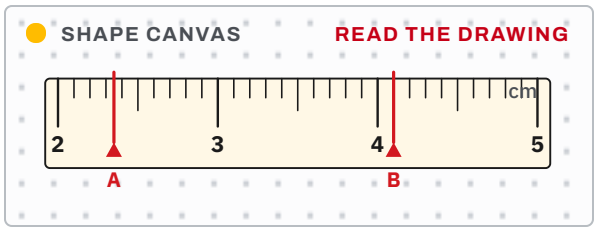
6 Write each size as a fraction in lowest terms.



WORKING

Answer _____

7 Read marks A and B. How far apart are they?



WORKING

Answer _____

6 Percent, Ratio and Proportion

EXERCISES

Practice with percents, decimals and fractions, percent of a number, allowances, increases and decreases, ratios, rates, proportions and scale drawings.

EXERCISE 6A

Skills

One skill at a time

Name _____

Date _____

1 Write 64% as a decimal.

WORKING

Answer _____

2 Write 3.5% as a decimal.

WORKING

Answer _____

3 Write 0.7 as a percent.

WORKING

Answer _____

4 Write 2.4 as a percent.

WORKING

Answer _____

5 Write 45% as a fraction in lowest terms.

WORKING

Answer _____

6 Write $\frac{7}{8}$ as a percent.

WORKING

Answer _____

7 Find 30% of 260.

WORKING

Answer _____

8 Find 6.5% of 4,200.

WORKING

Answer _____

9 27 is what percent of 360?

WORKING

Answer _____

10 42 is 35% of what number?

WORKING

Answer _____

11 Increase 340 by 15%.

WORKING

Answer _____

12 Decrease 1,250 by 8%.

WORKING

Answer _____

13 A value goes from 48 to 60. What is the percent increase?

WORKING

Answer _____

14 Write 36 : 84 in lowest terms.

WORKING

Answer _____

15 Write the ratio of 250 mm to 1.5 m in lowest terms.

WORKING

Answer _____

16 Share 1,200 in the ratio 5 : 3.

WORKING

Answer _____

17 A tank fills with 840 L in 12 minutes. What is the rate in L per minute?

WORKING

Answer _____

18 Solve $7/x = 21/36$.

WORKING

Answer _____

19 At a scale of 1 : 40, a line measures 85 mm. How long is the real length?

WORKING

Answer _____

20 A 14 mm part is drawn at 3 : 1. How long is it on the drawing?

WORKING

Answer _____

EXERCISE 6B

Shop problems

Fabrication, pipe trades and millwright jobs

Name _____

Date _____

1 A pipe run needs 64 m of pipe. Add a 12% waste allowance. The pipe on this job comes in 6.4 m lengths. How many metres do you need, and how many lengths do you order?

WORKING

Answer _____

2 A fabrication job needs 18 sheets of 4 ft × 8 ft plate. Add a 15% allowance and round up to whole sheets.

WORKING

Answer _____

3 A bearing lists at $86.50. Your shop gets a 12% trade discount. What does one bearing cost? What do 6 cost? Round to the cent.

WORKING

Answer _____

4 A conveyor belt moves 18 m in 45 s. What is its rate in m per s? At that rate, how long does a part take to travel 50 m?

WORKING

Answer _____

5 Of 85 welds tested, 6 are rejected. What percent are rejected, to two decimal places?

WORKING

Answer _____

6 Cutting 14 brackets uses 9.8 m of angle iron. How much angle iron do 35 brackets use?

WORKING

Answer _____

7 A two-part epoxy is mixed 5 : 2 by volume. You need 1.4 L of mixed epoxy. How much of each part do you measure?

WORKING

Answer _____

8 A drawing is marked 1/8″ = 1′-0″. A line measures 5 3/8 in. How long is the real length?

WORKING

Answer _____

9 Pipe insulation goes from $3.20 to $3.68 per foot. What is the percent increase?

WORKING

Answer _____

10 A drive pulley is 300 mm across and the driven pulley is 120 mm across. Write the ratio of the drive pulley to the driven pulley in lowest terms.

WORKING

Answer _____

EXERCISE 6C

Shape Canvas

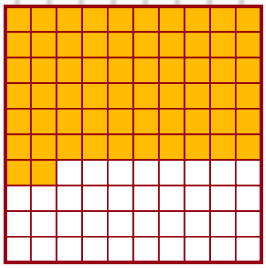
Find each red ?

Name _____

Date _____

1 Write the shaded part as a percent, a decimal and a fraction in lowest terms.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

2 The width is 35% of the length. Find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

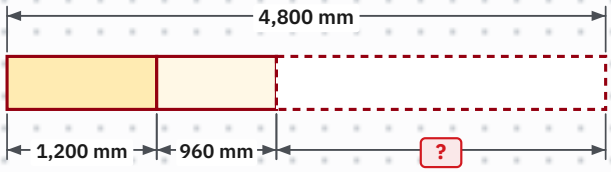


WORKING

Answer _____

3 Two pieces are cut from a 4,800 mm bar. Find the offcut. What percent of the bar is the offcut?

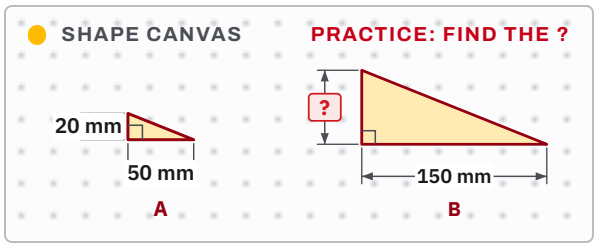
SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

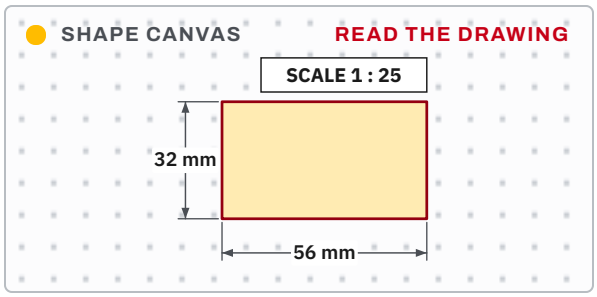
4 Triangle B is triangle A made larger. Find the height of B.



WORKING

Answer _____

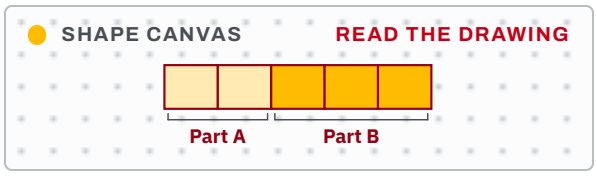
5 The sizes are measured on a 1 : 25 drawing. Find the real length and width.



WORKING

Answer _____

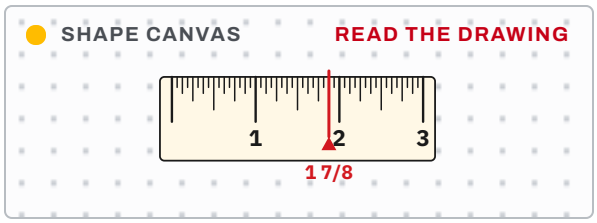
6 Part A and Part B are mixed as shown. How much of each makes 1.5 L?



WORKING

Answer _____

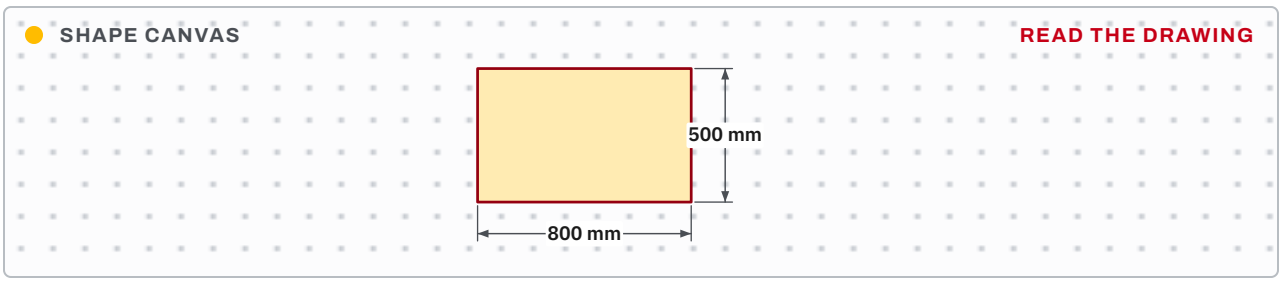
7 A drawing is marked 1/2″ = 1′-0″. Find the real length of the line in feet and inches.



WORKING

Answer _____

8 Change one value: a plate 800 mm × 500 mm is drawn at 1 : 10. Find the drawing sizes. Then find them at 1 : 5. What happens to every line?



WORKING

Answer _____

7 Measurement and Unit Conversion

EXERCISES

Practice with metric prefixes, feet and inches, exact conversion factors, area and volume units, litres and gallons, and temperature. Use 1 in = 25.4 mm, 1 ft = 0.3048 m, 1 lb = 0.453 592 37 kg, 1 US gal = 3.785 411 784 L and 1 imp gal = 4.546 09 L.

EXERCISE 7A

Skills

One conversion at a time

Name _____

Date _____

1 Convert 4.75 m to mm.

WORKING

Answer _____

2 Convert 625 mm to m.

WORKING

Answer _____

3 Convert 38 cm to mm.

WORKING

Answer _____

4 Convert 3,250 mL to L.

WORKING

Answer _____

5 Convert 2.4 t to kg.

WORKING

Answer _____

6 Convert 750 g to kg.

WORKING

Answer _____

7 Write 8′ 5″ in inches.

WORKING

Answer _____

8 Write 9′ 7 3/4″ in inches.

WORKING

Answer _____

9 Write 94 in in feet and inches.

WORKING

Answer _____

10 Write 157 1/2 in in feet and inches.

WORKING

Answer _____

11 Convert 5/8 in to mm.

WORKING

Answer _____

12 Convert 2 1/4 in to mm.

WORKING

Answer _____

13 Convert 950 mm to inches, to the nearest 1/16 in.

WORKING

Answer _____

14 Convert 15 ft to m.

WORKING

Answer _____

15 Convert 75 lb to kg, to two decimal places.

WORKING

Answer _____

16 How many feet are in 6 yd? How many metres?

WORKING

Answer _____

17 Convert 2.5 m^2 to mm^2.

WORKING

Answer _____

18 Convert 6 ft^2 to in^2.

WORKING

Answer _____

19 Convert 25 in^2 to mm^2.

WORKING

Answer _____

20 A tank is 800 mm × 400 mm × 300 mm inside. How many litres does it hold?

WORKING

Answer _____

21 Convert 20 US gal to L, to two decimal places.

WORKING

Answer _____

22 Convert 25 imp gal to L, to two decimal places.

WORKING

Answer _____

23 Convert 35 °C to °F.

WORKING

Answer _____

24 Convert −4 °F to °C.

WORKING

Answer _____

EXERCISE 7B

Shop problems

Metric drawings, imperial tools and mixed units

Name _____

Date _____

1 A metric drawing gives a length of 2,235 mm. Your tape reads in feet and inches. What do you mark, to the nearest 1/16 in?

WORKING

Answer _____

2 Three pieces of pipe are 4′ 6 1/2″, 7′ 3 3/4″ and 2′ 11″ long. Find the total length in feet and inches and in millimetres.

WORKING

Answer _____

3 12 boxes of fittings weigh 22 lb each. What is the total mass in kilograms, to two decimal places?

WORKING

Answer _____

4 A shop floor is 6.2 m by 4.5 m. Find its area in m² and in ft², to two decimal places. The coating label says one US gallon covers 350 ft². How many whole gallons do you buy?

WORKING

Answer _____

5 A hydraulic reservoir is 900 mm × 600 mm × 450 mm inside. How many litres does it hold? How many US gallons, to two decimal places?

WORKING

Answer _____

6 An older air receiver is marked 30 imp gal. What is that in US gallons, to two decimal places?

WORKING

Answer _____

7 A welding procedure gives a preheat of 200 °F and a maximum temperature between passes of 120 °C. Write the preheat in °C, to one decimal place, and the maximum in °F.

WORKING

Answer _____

8 A shaft warms up by 35 °C while running. How many °F is that rise?

WORKING

Answer _____

9 A flat bar is 1/2″ × 6″ × 30″. Find its volume in in³, then in cm³ to two decimal places.

WORKING

Answer _____

10 A metric sheet is 1,220 mm × 2,440 mm. Find its area in m², then in ft² to two decimal places.

WORKING

Answer _____

EXERCISE 7C

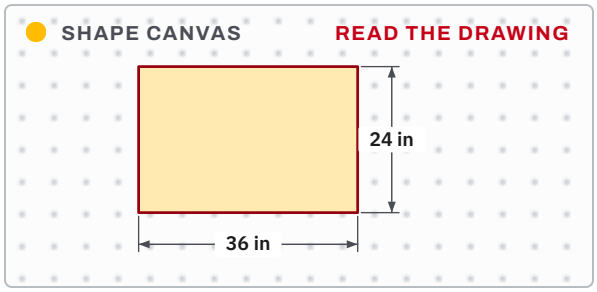
Shape Canvas

Convert the drawings and find each red ?

Name _____

Date _____

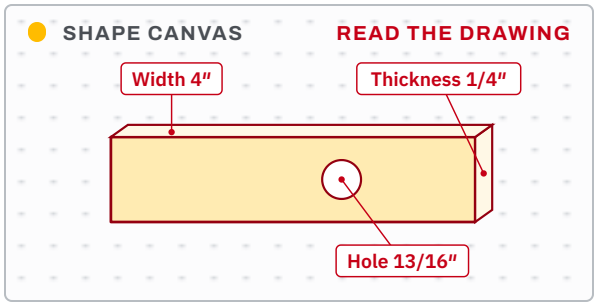
1 Write both sizes of the sheet in millimetres.



WORKING

Answer _____

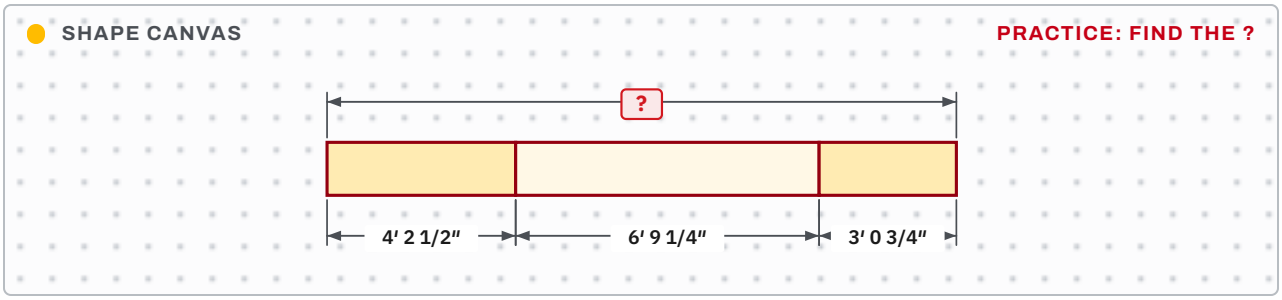
2 Convert each size on the plate to millimetres.



WORKING

Answer _____

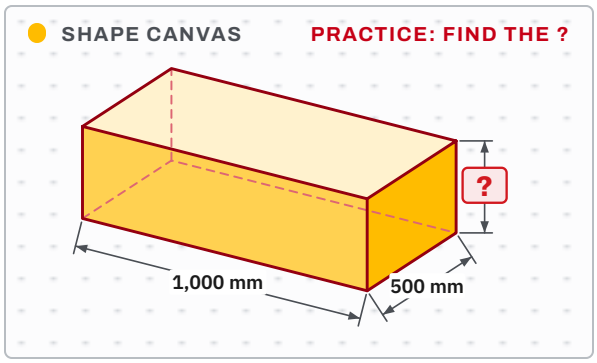
3 Find the overall length in feet and inches and in millimetres.



WORKING

Answer _____

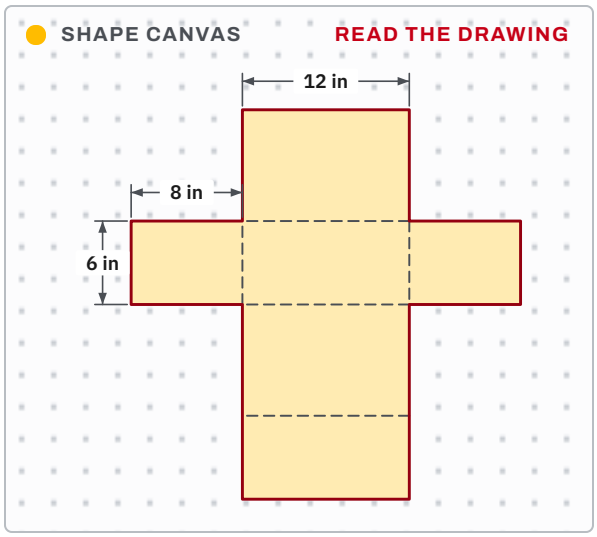
4 The tank holds 150 L. Find the hidden height.



WORKING

Answer _____

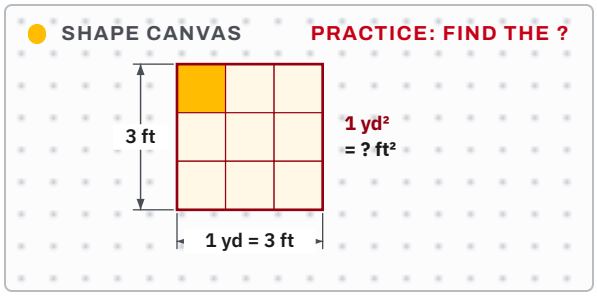
5 Unfold it: the net of a closed box in inches. Find the total area in in² and in ft².



WORKING

Answer _____

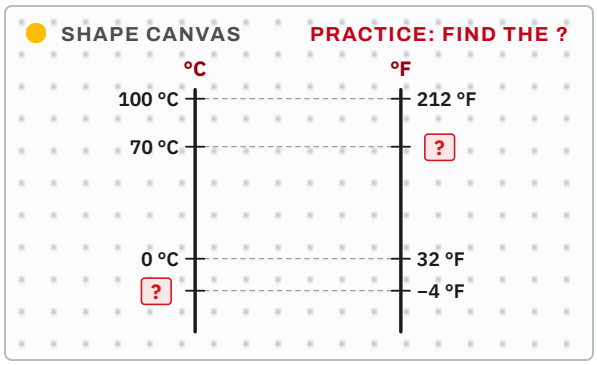
6 A square yard split into square feet. How many ft² are in 1 yd²? How many in²?



WORKING

Answer _____

7 Find each red ? on the two scales.



WORKING

Answer _____

8 Change one value: a cube is 50 mm on each side. How many litres does it hold? Each side is doubled to 100 mm. How many litres now, and how many times as much?

WORKING

Answer _____

8 Formulas and Transposing

EXERCISES

Practice reading formulas, substituting values, solving with inverse operations and transposing formulas, including formulas with squares and square roots. Check every answer by substituting it back.

EXERCISE 8A

Skills

Name the parts, substitute, solve and transpose

Name _____

Date _____

1 In $V = \pi \times r^2 \times h$, name the subject, the variables and the constant.

WORKING

Answer _____

2 Rewrite $3ab$ and πr^2 with × signs.

WORKING

Answer _____

3 Substitute: $A = l \times w$ with $l = 3.6$ m and $w = 2.5$ m.

WORKING

Answer _____

4 Substitute: $P = 2 \times (l + w)$ with $l = 1{,}250$ mm and $w = 875$ mm.

WORKING

Answer _____

5 Substitute: $C = \pi \times D$ with $D = 450$ mm. Round to two decimal places.

WORKING

Answer _____

6 Substitute: $A = \pi \times r^2$ with $r = 35$ mm. Round to two decimal places.

WORKING

Answer _____

7 Substitute: $V = l \times w \times h$ with $l = 1.2$ m, $w = 0.9$ m and $h = 0.75$ m.

WORKING

Answer _____

8 Substitute: $t_F = 1.8 \times t_C + 32$ with $t_C = -10$ °C.

WORKING

Answer _____

9 Solve $x + 175 = 640$.

WORKING

Answer _____

10 Solve $x - 38 = 112$.

WORKING

Answer _____

11 Solve $8 \times x = 2{,}000$.

WORKING

Answer _____

12 Solve $x \div 6 = 45$.

WORKING

Answer _____

13 Solve $15 \times x = 1{,}350$.

WORKING

Answer _____

14 Solve $x - 2.75 = 9.5$.

WORKING

Answer _____

15 Make l the subject of $A = l \times w$.

WORKING

Answer _____

16 Make D the subject of $C = \pi \times D$.

WORKING

Answer _____

17 Make w the subject of $V = l \times w \times h$.

WORKING

Answer _____

18 Make s the subject of $P = 4 \times s$.

WORKING

Answer _____

19 Make l the subject of $P = 2 \times (l + w)$.

WORKING

Answer _____

20 Make t_C the subject of $t_F = 1.8 \times t_C + 32$.

WORKING

Answer _____

21 Make b the subject of $A = b \times h \div 2$.

WORKING

Answer _____

22 Make h the subject of $T = a + r \times h$.

WORKING

Answer _____

23 Make s the subject of $A = s^2$.

WORKING

Answer _____

24 Make r the subject of $A = \pi \times r^2$.

WORKING

Answer _____

25 Make h the subject of $V = \pi \times r^2 \times h$.

WORKING

Answer _____

26 Make n the subject of $L = n \times \pi \times D$, where n is a number of turns.

WORKING

Answer _____

EXERCISE 8B

Shop problems

Transpose first, then substitute and check

Name _____

Date _____

1 A floor plate must cover 0.54 m² and is 0.6 m wide. How long is it?

WORKING

Answer _____

2 A rectangular tank base is 1.5 m by 1.0 m. The tank must hold 1.8 m³. How tall is it?

WORKING

Answer _____

3 A round gasket must have an area of 12,000 mm². Find its diameter to two decimal places.

WORKING

Answer _____

4 A welded frame uses 6,400 mm of tube all the way around. It is 1,250 mm wide. How long is it?

WORKING

Answer _____

5 A welding procedure calls for a preheat of 250 °F. Change it to °C, to two decimal places.

WORKING

Answer _____

6 A square base plate has an area of 1,296 in². Find the side in inches and in feet.

WORKING

Answer _____

7 A round tank has a radius of 0.75 m and must hold 4 m³. Use $V = \pi \times r^2 \times h$ to find its height, to two decimal places.

WORKING

Answer _____

8 A rental invoice uses $T = 45 + 18.50 \times d$, with T in dollars and d in days. The total is $285.50. For how many days was the welder rented?

WORKING

Answer _____

9 A tape wrapped once around a pipe reads $52\frac{3}{8}$ in. Find the diameter to the nearest 1/16 in.

WORKING

Answer _____

10 A triangular gusset must have an area of 15,000 mm². Its height is 150 mm. Use $A = b \times h \div 2$ to find the base.

WORKING

Answer _____

11 A drawing gives a disc an area of 0.25 m². Find its radius to three decimal places.

WORKING

Answer _____

12 A shelf is 1.8 m long and 0.6 m deep, so its area is 1.08 m². A part laid flat is 1.6 m by 0.65 m. Find the area of the part. Does the part fit on the shelf? Explain using areas and sizes.

WORKING

Answer _____

13 A sheet is 48 in wide and has an area of 4,608 in². Find its length in inches and in feet.

WORKING

Answer _____

14 A cart wheel is 18 in in diameter. Use $n = L \div (\pi \times D)$ to find how many turns it makes when the cart moves 150 ft. Round to two decimal places.

WORKING

Answer _____

15 Find the mistake. A learner transposes $P = 2 \times (l + w)$ to $l = P - 2 \times w$ and uses P = 4,000 mm and w = 700 mm. Find the learner's answer, check it by substituting back, and find the correct length.

WORKING

Answer _____

EXERCISE 8C

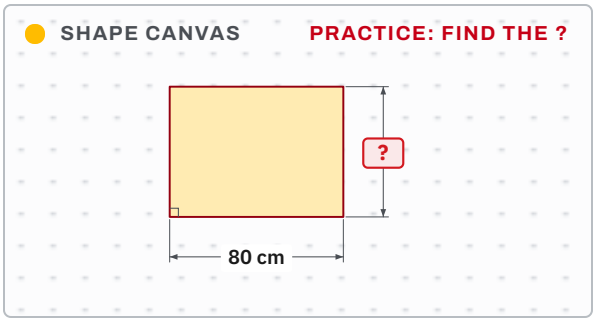
Shape Canvas

Use the value you know to find the ?

Name _____

Date _____

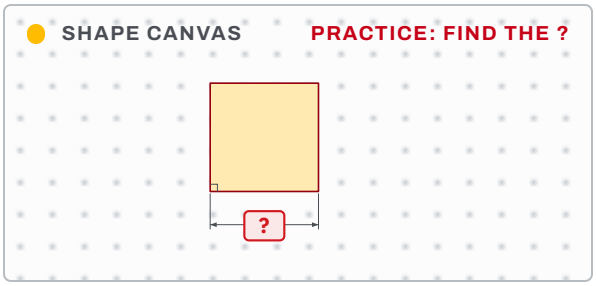
1 The rectangle has an area of 4,800 cm². Find the width.



WORKING

Answer _____

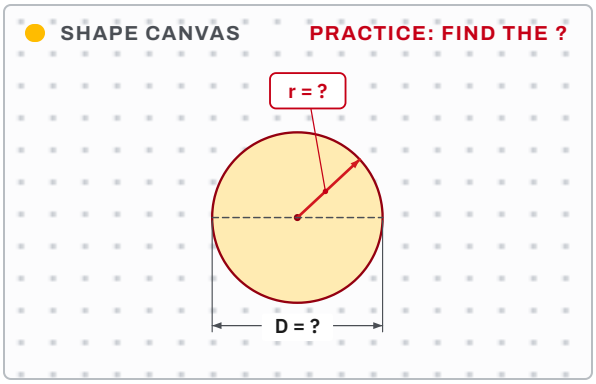
2 The square has an area of 3,136 mm². Find the side.



WORKING

Answer _____

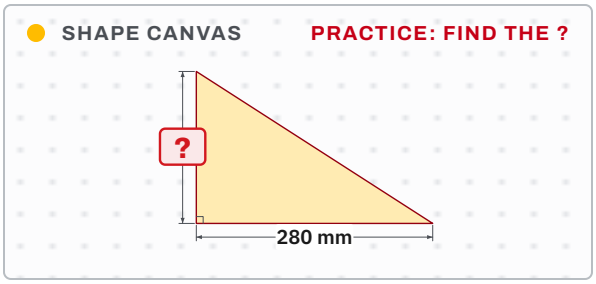
3 The circle has an area of 8,000 mm². Find r and D to two decimal places.



WORKING

Answer _____

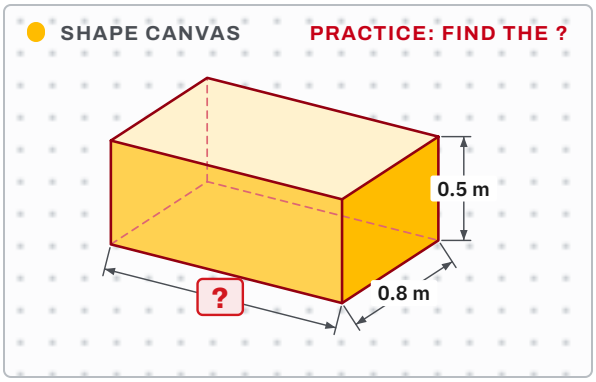
4 The triangle has an area of 25,200 mm². Find the height.



WORKING

Answer _____

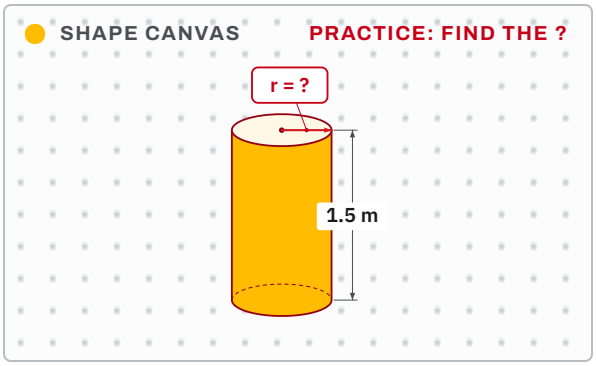
5 The box holds 0.48 m³. Find its length.



WORKING

Answer _____

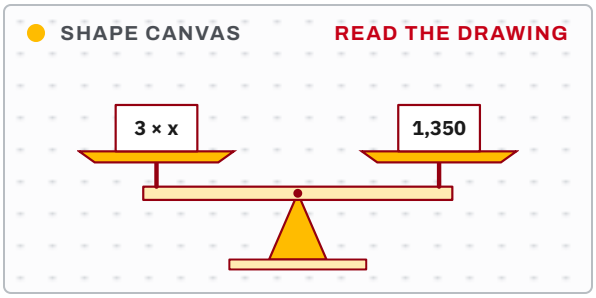
6 The tank holds 0.9 m³. Find its radius to two decimal places.



WORKING

Answer _____

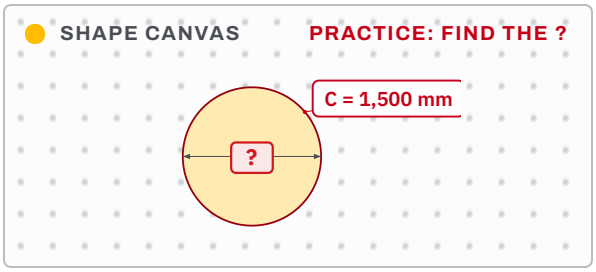
7 The scale is level. Find x.



WORKING

Answer _____

8 A tape around the pipe reads 1,500 mm. Find D to two decimal places.



WORKING

Answer _____

9 Change one value: the square in problem 2 is given an area 9 times as big. Find the new side.

WORKING

Answer _____

10 Change one value: the circle in problem 3 is given an area 4 times as big, 32,000 mm². Find the new diameter to two decimal places. What happened to it?

WORKING

Answer _____

9 Perimeter and Circumference

EXERCISES

Practice finding the distance around straight-sided shapes and circles, the stretch-out of pipes and ducts, arcs and composite shapes. The distance around a circle is always π × D.

EXERCISE 9A

Skills

Perimeter, parts of a circle, circumference and arcs

Name _____

Date _____

1 Find the perimeter of a rectangle 2,400 mm by 1,150 mm.

WORKING

Answer _____

2 Find the perimeter of a square with sides of 75 cm.

WORKING

Answer _____

3 A triangle has sides of 0.8 m, 1.1 m and 1.3 m. Find the perimeter.

WORKING

Answer _____

4 A regular pentagon has sides of 90 mm. Find the perimeter.

WORKING

Answer _____

5 A regular octagon has sides of 45 in. Find the perimeter in feet.

WORKING

Answer _____

6 Find the perimeter of a rectangle $15\frac{3}{4}$ in by $9\frac{5}{8}$ in.

WORKING

Answer _____

7 Find the perimeter of a rectangle 9 ft 3 in by 5 ft 10 in, in feet and inches.

WORKING

Answer _____

8 A rectangle has a perimeter of 3,800 mm and a length of 1,200 mm. Find the width.

WORKING

Answer _____

9 A circle has a diameter of 273 mm. Find the radius.

WORKING

Answer _____

10 A circle has a radius of 82.5 mm. Find the diameter.

WORKING

Answer _____

11 D = 380 mm. Find C to two decimal places.

WORKING

Answer _____

12 r = 95 mm. Find C to two decimal places.

WORKING

Answer _____

13 C = 1,500 mm. Find D to two decimal places.

WORKING

Answer _____

14 D = 14 in. Find C to the nearest 1/16 in.

WORKING

Answer _____

15 Find the length of a 45° arc of a circle with D = 800 mm. Round to two decimal places.

WORKING

Answer _____

16 Find the length of a 150° arc of a circle with r = 600 mm. Round to two decimal places.

WORKING

Answer _____

17 Find the curved edge of a half circle with D = 640 mm, to two decimal places.

WORKING

Answer _____

18 A circle has its diameter cut in half. What happens to its circumference?

WORKING

Answer _____

EXERCISE 9B

Shop problems

Fabrication, pipe trades and millwright

Name _____

Date _____

1 A frame of angle iron is 2 ft 6 in by 1 ft 9 in. How much angle iron goes around it? Give feet and inches.

WORKING

Answer _____

2 A steel band goes once around a tank 1,850 mm in diameter, with 150 mm extra for the overlap. Find the band length to the nearest millimetre.

WORKING

Answer _____

3 A round duct is 355 mm in diameter. The drawing adds 25 mm to the stretch-out for the seam. Find the cut length to the nearest millimetre.

WORKING

Answer _____

4 An insulation jacket wraps a pipe with an outside diameter of $10\frac{3}{4}$ in. Find the stretch-out to the nearest 1/16 in.

WORKING

Answer _____

5 A pulley 250 mm in diameter turns 1,450 times. How far does a point on its rim travel, in metres to two decimal places?

WORKING

Answer _____

6 A cart wheel is 24 in in diameter. How many turns does it make in 100 ft? Round to two decimal places.

WORKING

Answer _____

7 A fillet weld runs once around a pipe with an outside diameter of 168.3 mm. How long is the weld, to two decimal places?

WORKING

Answer _____

8 A tape wrapped around a round column reads 2,042 mm. Find the diameter to the nearest millimetre.

WORKING

Answer _____

9 Four guard frames, each 1,200 mm by 800 mm, are cut from 6,000 mm lengths of flat bar. Ignore saw cuts. How much bar is needed, and how many 6,000 mm lengths must be ordered?

WORKING

Answer _____

10 A handrail follows a 90° arc of a circle with a 1,500 mm radius. How long is the rail, to two decimal places?

WORKING

Answer _____

11 A belt runs around two pulleys, each 300 mm in diameter, with centres 1,100 mm apart. The pulleys are moved 150 mm farther apart. Find the belt length before and after. How much longer is the new belt? Round to two decimal places.

WORKING

Answer _____

12 A round tank lid is 1,220 mm in diameter. A rubber gasket goes once around its edge, with 40 mm extra for the joint. Find the gasket length to two decimal places.

WORKING

Answer _____

13 A rectangular duct is 600 mm by 400 mm. A strip of insulation wraps once around it with a 50 mm overlap. How long must the strip be?

WORKING

Answer _____

EXERCISE 9C

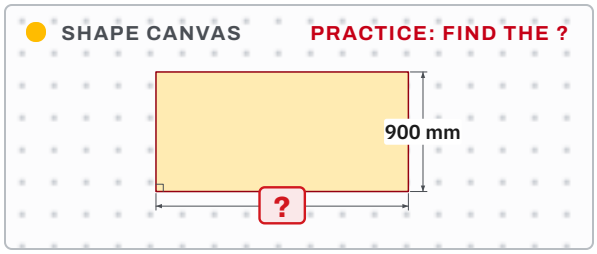
Shape Canvas

Find every ?, break it down, unfold it. Round answers to two decimal places.

Name _____

Date _____

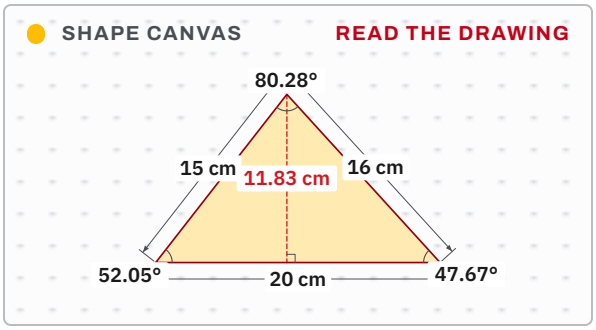
1 The perimeter is 5,600 mm. Find the length.



WORKING

Answer _____

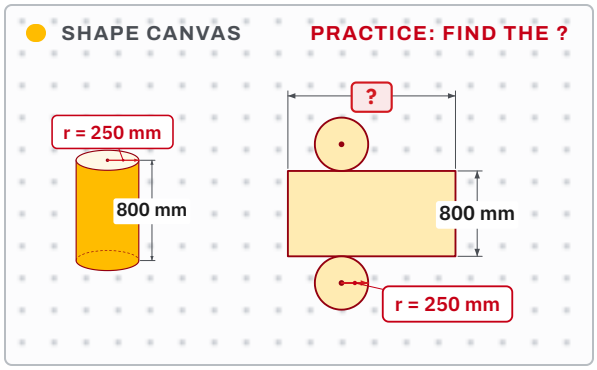
2 Find the perimeter. The dashed line is the height, not an edge.



WORKING

Answer _____

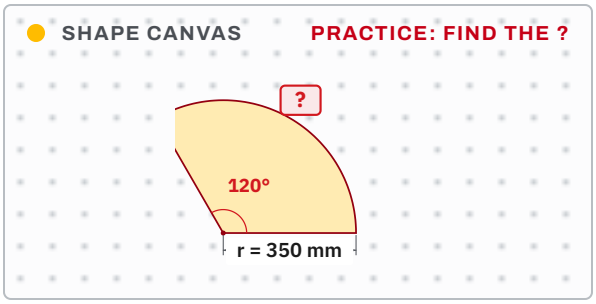
3 Unfold it. Find the stretch-out marked ?.



WORKING

Answer _____

4 Find the length of the arc marked ?.

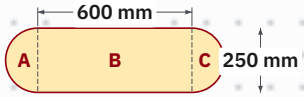


WORKING

Answer _____

5 Break it down: the plate has a half circle on each end. Find its perimeter.

SHAPE CANVAS READ THE DRAWING

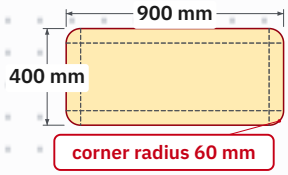


WORKING

Answer _____

6 Each corner is a quarter circle. Find the perimeter of the plate.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

7 The belt is 2,400 mm long. Find the centre distance marked ?.

SHAPE CANVAS PRACTICE: FIND THE ?

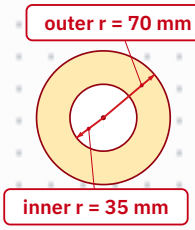


WORKING

Answer _____

8 Change one value: the outside of the washer is kept, but the hole is made 10 mm bigger across. Find the new total length of both edges.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

10 Area

EXERCISES

Practice with square units, the area formulas for rectangles, triangles, parallelograms, trapezoids, circles and rings, composite shapes and surface area from nets. Use the π key for circles.

EXERCISE 10A

Skills

One skill at a time

Name _____

Date _____

1 A plate drawn on a 1 cm grid covers 5 rows of 9 squares. What is its area?

WORKING

Answer _____

2 Write 1,500,000 mm² in m².

WORKING

Answer _____

3 Write 0.08 m² in mm².

WORKING

Answer _____

4 Write 6 ft² in in². Then write 432 in² in ft².

WORKING

Answer _____

5 Find the area of a rectangle 840 mm by 350 mm.

WORKING

Answer _____

6 Find the area of a square with 1.6 m sides.

WORKING

Answer _____

7 Triangle: base 520 mm, height 270 mm. Find the area.

WORKING

Answer _____

8 Triangle: base 15 in, height 7 in. Find the area.

WORKING

Answer _____

9 Parallelogram: base 0.75 m, slanted side 0.5 m, height 0.4 m. Find the area.

WORKING

Answer _____

10 Trapezoid: parallel sides 250 mm and 410 mm, height 180 mm. Find the area.

WORKING

Answer _____

11 Circle, radius 45 mm. Find the area to two decimal places.

WORKING

Answer _____

12 Circle, diameter 30 in. Find the area to two decimal places.

WORKING

Answer _____

13 Half circle, radius 1.2 m. Find the area to two decimal places.

WORKING

Answer _____

14 Quarter circle, radius 500 mm. Find the area to the nearest mm^2.

WORKING

Answer _____

15 Ring: outside diameter 120 mm, inside diameter 80 mm. Find the area to two decimal places.

WORKING

Answer _____

16 A ring has an outside radius of 60 mm and an inside radius of 40 mm. A student finds its area with $\pi \times (60 - 40)^2$. Explain the mistake.

WORKING

Answer _____

17 A triangle has an area of 54 in^2 and a base of 12 in. Find its height.

WORKING

Answer _____

18 Write 2.5 m^2 in cm^2. (1 m^2 = 10,000 cm^2)

WORKING

Answer _____

EXERCISE 10B

Shop problems

Area at work, metric and imperial

Name _____

Date _____

1 A mezzanine floor is 7.2 m by 3.6 m. It is covered with checker plate sheets that are 1.2 m by 2.4 m, laid edge to edge with no waste. Find the floor area and the number of sheets.

WORKING

Answer _____

2 Four plates, each 2.5 m by 1.6 m, are primed on both sides. The primer label says 1 L covers 7 m². Primer is sold in 1 L tins. How many tins do you buy? Give the litres to two decimal places.

WORKING

Answer _____

3 A full-face gasket is a ring with an outside diameter of 340 mm and an inside diameter of 220 mm. Find the area of gasket material to two decimal places.

WORKING

Answer _____

4 A job needs 24 triangular gussets, each with a base of 16 in and a height of 12 in. Find the total area in in² and in ft².

WORKING

Answer _____

5 A base plate is 450 mm square with four 22 mm diameter bolt holes. Find the area of steel left, to two decimal places.

WORKING

Answer _____

6 A hopper has four equal trapezoid sides. Each has parallel sides of 1,500 mm and 300 mm, and a height of 900 mm. Find the total area of the four sides in m².

WORKING

Answer _____

7 A duct 18 in by 12 in and 96 in long is open at both ends. Ignore seams. Find the area of sheet in ft².

WORKING

Answer _____

8 A closed cylindrical tank is 1.2 m across and 2 m high. Find its surface area in m² to two decimal places.

WORKING

Answer _____

9 A machine guard panel is 900 mm by 600 mm with a half-circle notch of radius 150 mm cut from one long edge. Find the area of the panel to two decimal places.

WORKING

Answer _____

10 A sheet metal jacket wraps once around a pipe with an outside diameter of 168 mm. The jacket is 3,000 mm long. Ignore laps. Find the area of the jacket in m² to two decimal places.

WORKING

Answer _____

11 A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

WORKING

Answer _____

12 A shop floor is 14 ft by 9 ft 6 in. Find its area in ft².

WORKING

Answer _____

EXERCISE 10C

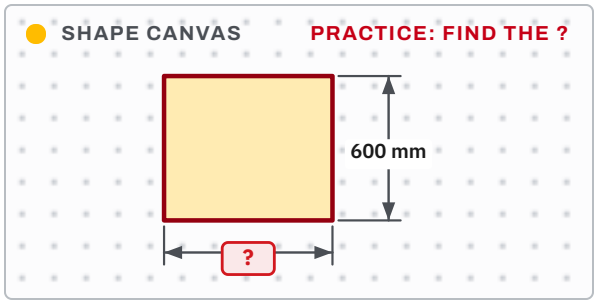
Shape Canvas

Hidden sizes, break it down, unfold it, change one value

Name _____

Date _____

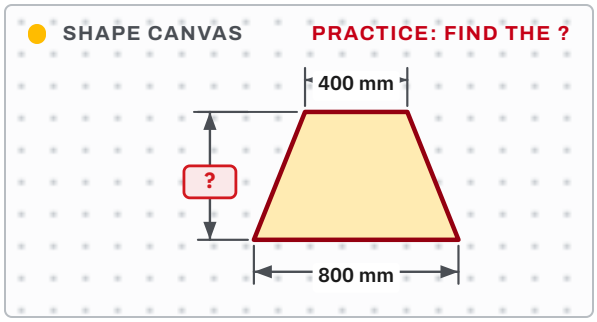
1 The plate has an area of 0.42 m². Find the hidden length.



WORKING

Answer _____

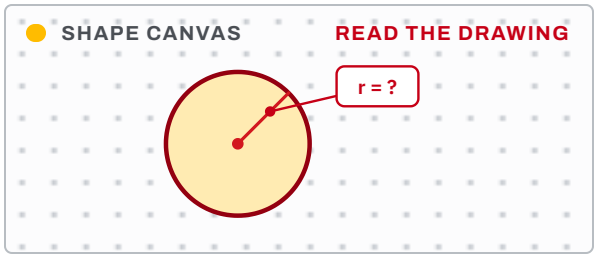
2 The trapezoid has an area of 300,000 mm². Find the hidden height.



WORKING

Answer _____

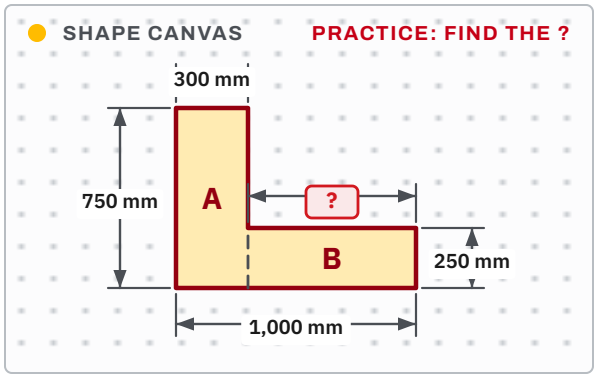
3 The disc has an area of 31,416 mm². Find the hidden radius to the nearest mm.



WORKING

Answer _____

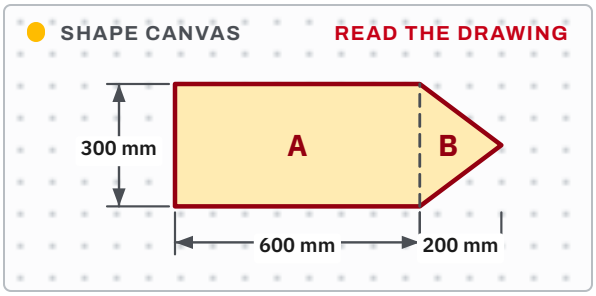
4 Find the hidden length of the foot B, then the area of the L-plate.



WORKING

Answer _____

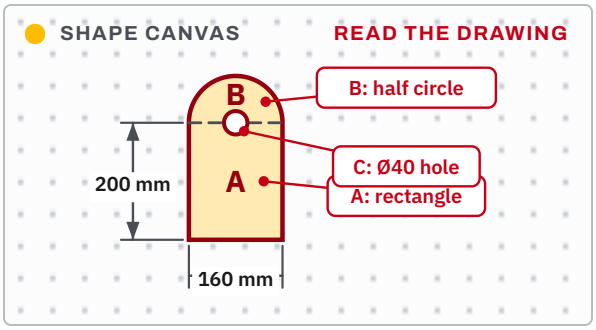
5 Find the area of the plate with the pointed end.



WORKING

Answer _____

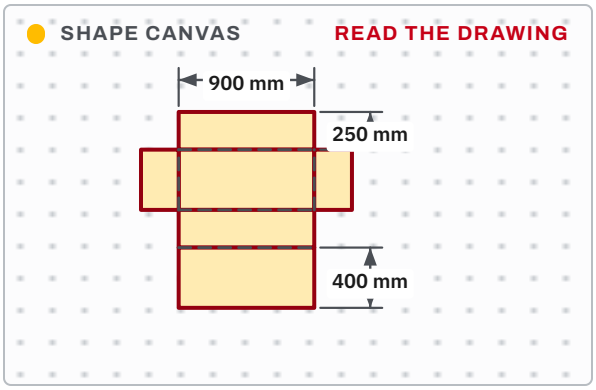
6 Find the area of the lug, A + B − C, to two decimal places.



WORKING

Answer _____

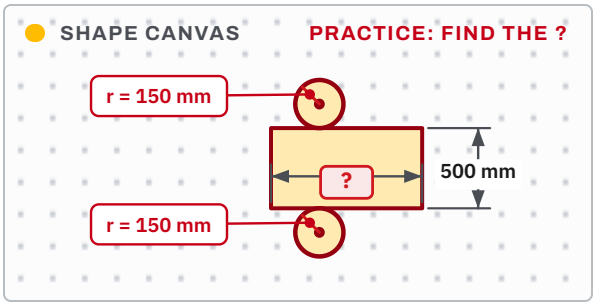
7 Unfold it: this is the net of a closed box. Find the surface area of the box in mm² and m².



WORKING

Answer _____

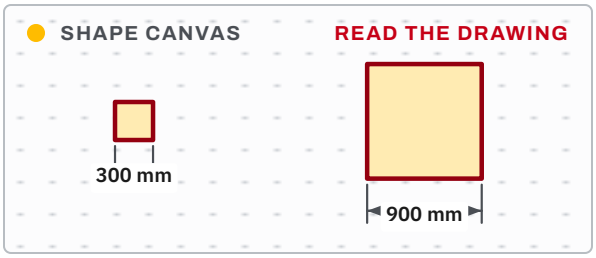
8 This net makes a closed cylinder with a radius of 150 mm. Find the hidden stretch-out and the surface area, to two decimal places.



WORKING

Answer _____

9 Change one value: the 300 mm side of a square plate is made 3 times longer. Find both areas. How many times bigger is the new area?



WORKING

Answer _____

11 Volume and Capacity

EXERCISES

Practice with cubic units, the volume of prisms, cylinders, cones, pyramids and spheres, composite solids, and capacity in litres and US gallons (1 US gal = 231 in³ = 3.785411784 L).

EXERCISE 11A

Skills

One skill at a time

Name _____

Date _____

1 A block of 1 cm cubes is 6 long, 4 wide and 3 high. Find its volume.

WORKING

Answer _____

2 Write 3 cm³ in mm³.

WORKING

Answer _____

3 Write 2,500,000,000 mm³ in m³.

WORKING

Answer _____

4 Write 0.45 m³ in litres.

WORKING

Answer _____

5 Write 3,456 in³ in ft³.

WORKING

Answer _____

6 Write 2,000,000 mm³ in litres.

WORKING

Answer _____

7 Box 700 mm by 400 mm by 250 mm. Find the volume in m³.

WORKING

Answer _____

8 Cube with 0.6 m edges. Find the volume in m³.

WORKING

Answer _____

9 Triangular prism: end base 300 mm, end height 200 mm, length 1,500 mm. Find the volume in litres.

WORKING

Answer _____

10 Write 1,155 in³ in US gallons.

WORKING

Answer _____

11 Write 75 US gal in litres, to two decimal places.

WORKING

Answer _____

12 Write 500 L in US gallons, to two decimal places.

WORKING

Answer _____

13 Cylinder: radius 150 mm, height 600 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

14 Cone: radius 200 mm, height 450 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

15 Square pyramid: base 900 mm by 900 mm, height 600 mm. Capacity in litres?

WORKING

Answer _____

16 Sphere: radius 250 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

17 Hemisphere: radius 0.4 m. Volume in m^3, to three decimal places?

WORKING

Answer _____

18 Pipe with an inside diameter of 80 mm. Litres per metre, to two decimal places?

WORKING

Answer _____

19 A box holds 0.24 m^3. Its base is 0.8 m by 0.5 m. Find its height.

WORKING

Answer _____

20 How many US gallons are in 1 ft^3 (1,728 in^3), to two decimal places?

WORKING

Answer _____

21 Which holds more, a 40 L drum or a 10 US gal can? Show the litres to two decimal places.

WORKING

Answer _____

22 Cylinder: radius 0.2 m, height 0.5 m. Volume in m^3, to three decimal places?

WORKING

Answer _____

EXERCISE 11B

Shop problems

Tanks, pipes and parts, metric and imperial

Name _____

Date _____

1 A hydraulic reservoir is 800 mm by 500 mm by 450 mm inside. How many litres does it hold? How many 20 L pails of oil fill it?

WORKING

Answer _____

2 A water heater tank is a cylinder 18 in across and 48 in tall inside. Find its capacity in US gallons, to two decimal places.

WORKING

Answer _____

3 Before a pressure test, 75 m of pipe with an inside diameter of 52 mm is filled with water. How many litres, to two decimal places?

WORKING

Answer _____

4 A system has 40 m of pipe with a 100 mm inside diameter and 25 m with a 50 mm inside diameter. Find the total litres, to two decimal places.

WORKING

Answer _____

5 A sump is 1.5 m by 1.0 m. The water in it is 0.6 m deep. A pump removes 45 L per minute. How many litres are in the sump, and how long does pumping take?

WORKING

Answer _____

6 A hopper is a cylinder 1.2 m across and 1.0 m high on a cone 0.8 m high. Find its capacity in litres, to two decimal places.

WORKING

Answer _____

7 A drum is 22.5 in across and 33.5 in tall inside. Find its capacity in US gallons, to two decimal places.

WORKING

Answer _____

8 A tank is a cylinder with a hemisphere on each end. The cylinder part is 2 m long. The radius is 0.5 m. Find its capacity in litres, to two decimal places.

WORKING

Answer _____

9 A vertical tank is 1 m across inside and holds water 1.4 m deep. How many litres are in it, to two decimal places?

WORKING

Answer _____

10 A solid round bar is 50 mm in diameter and 3 m long. Find its volume in cm^3, to two decimal places.

WORKING

Answer _____

11 A tube has an outside diameter of 114 mm, an inside diameter of 102 mm and a length of 6,000 mm. Find the volume of metal in cm^3, to two decimal places.

WORKING

Answer _____

12 A fuel day tank is 36 in by 24 in by 20 in inside. Find its capacity in US gallons to two decimal places. A heater burns 2.5 US gal per hour. How many hours will a full tank last, to the nearest hour?

WORKING

Answer _____

13 A blast-media hopper is an upside-down square pyramid. The top is 1,200 mm square and the depth is 900 mm. Find its capacity in litres.

WORKING

Answer _____

EXERCISE 11C

Shape Canvas

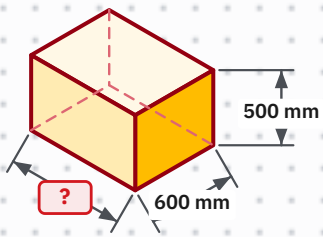
Hidden sizes, break it down, unfold it, change one value

Name _____

Date _____

1 The tank holds 240 L. Find the hidden length.

SHAPE CANVAS PRACTICE: FIND THE ?

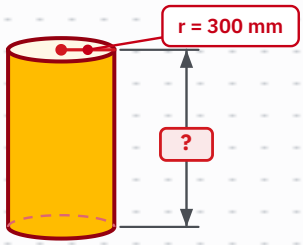


WORKING

Answer _____

2 The cylinder holds 282.74 L to two decimal places. Find the hidden height to the nearest mm.

SHAPE CANVAS PRACTICE: FIND THE ?

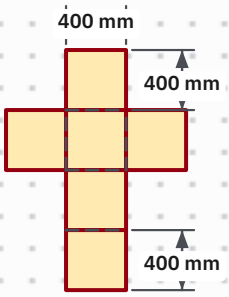


WORKING

Answer _____

3 Fold the net into a cube. Find its capacity in litres.

SHAPE CANVAS READ THE DRAWING

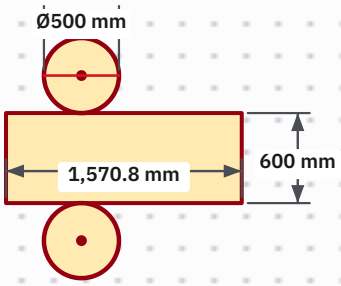


WORKING

Answer _____

4 Fold the net into a closed cylinder. Find its capacity in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING

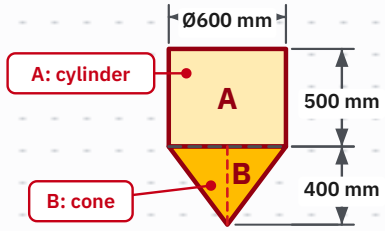


WORKING

Answer _____

5 Break it down: A is a cylinder 600 mm across and B is a cone. Find the capacity of the hopper, A + B, in litres to two decimal places.

SHAPE CANVAS READ THE DRAWING

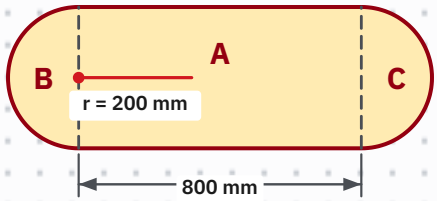


WORKING

Answer _____

6 The tank has a hemisphere on each end. Find its capacity in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING

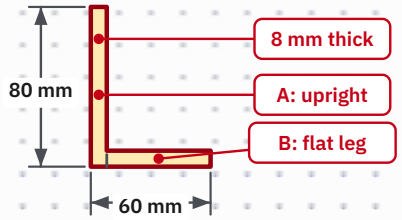


WORKING

Answer _____

7 Break down the L-shaped end of the angle bar into two rectangles. Find the end area and the volume in cm³.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

8 Change one value: a spherical float with a radius of 100 mm has its radius doubled. Find both capacities in litres, to two decimal places. How many times bigger is the new one?

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

9 Change one value: a cone has a radius of 300 mm and a height of 300 mm. The height is doubled. Find both capacities in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

12 Angles and Right Triangles

EXERCISES

Practice with degrees, angle names, missing angles in triangles, the Pythagorean theorem, the 3-4-5 check and diagonals.

EXERCISE 12A

Skills

Angles, triangles and the Pythagorean theorem, one skill at a time.

Name _____

Date _____

1 How many degrees are in one third of a turn?

WORKING

Answer _____

2 Nine holes are evenly spaced around a circle. What is the angle between neighbours?

WORKING

Answer _____

3 Name each angle: 12°, 90°, 179°, 95°.

WORKING

Answer _____

4 Two angles sit on a straight line. One is 64°. Find the other.

WORKING

Answer _____

5 Four angles go around a point: 80°, 95°, 110° and ?. Find the ?

WORKING

Answer _____

6 A triangle has angles of 52° and 71°. Find the third angle.

WORKING

Answer _____

7 One acute angle of a right triangle is 38°. Find the other.

WORKING

Answer _____

8 A triangle has two equal sides with 64° between them. Find the other two angles.

WORKING

Answer _____

9 A triangle has two equal angles of 35°. Find the third angle.

WORKING

Answer _____

10 Legs of 700 mm and 2,400 mm. Find the hypotenuse.

WORKING

Answer _____

11 Legs of 350 mm and 480 mm. Find the hypotenuse to two decimal places.

WORKING

Answer _____

12 Hypotenuse 1,700 mm, one leg 800 mm. Find the other leg.

WORKING

Answer _____

13 Hypotenuse 2.6 m, one leg 1.4 m. Find the other leg to two decimal places.

WORKING

Answer _____

14 Legs of 14″ and 19″. Find the hypotenuse to the nearest 1/16″.

WORKING

Answer _____

15 Do 8, 15 and 17 make a right triangle? Show the check.

WORKING

Answer _____

16 Do 6, 9 and 11 make a right triangle? Show the check.

WORKING

Answer _____

17 A rectangle is 1,100 mm by 600 mm. Find the diagonal to the nearest millimetre.

WORKING

Answer _____

18 A square has 450 mm sides. Find the diagonal to two decimal places.

WORKING

Answer _____

EXERCISE 12B

Shop problems

Layout, bracing and squaring in the shop. Metric and imperial.

Name _____

Date _____

1 A pump flange has 12 bolt holes evenly spaced on its bolt circle. What is the angle between neighbouring holes? What is the angle at the centre from hole 1 to hole 4?

WORKING

Answer _____

2 A flat bar is bent so that one side of the bend makes 142° with the straight part. What angle is on the other side of the bend? Is it acute or obtuse?

WORKING

Answer _____

3 A corner brace on a machine stand runs from a point 450 mm up the leg to a point 600 mm along the rail. Both points are measured from the inside corner. How long is the brace, centre line to centre line? Two braces are cut from one bar with two cuts, 3 mm kerf each. How much bar do they use?

WORKING

Answer _____

4 You square up a skid on the floor. You mark 1.2 m along one side and 1.6 m along the other. The tape reads 1.998 m between the marks. What should it read? Is the corner open or closed?

WORKING

Answer _____

5 A pipe rack frame is 3,000 mm by 1,800 mm. The diagonals measure 3,505 mm and 3,492 mm. What should each diagonal measure, to the nearest millimetre? Which two corners do you push toward each other?

WORKING

Answer _____

6 A sloped support covers 2,250 mm along the floor and rises 800 mm. How long is it, to two decimal places?

WORKING

Answer _____

7 A gate frame is 4′ 0″ wide and 6′ 6″ tall. Find the length of the diagonal brace, corner to corner, to the nearest 1/16″. Give it in feet and inches.

WORKING

Answer _____

8 A guard is bent from one bar into a right triangle. The two short sides are 18″ and 24″. How long is the bar before bending? Ignore the bends.

WORKING

Answer _____

9 A three-arm bracket has its arms evenly spaced around a hub. What is the angle between arms? If one arm is moved so one of the angles next to it is 100°, what are the other two angles?

WORKING

Answer _____

EXERCISE 12C

Shape Canvas

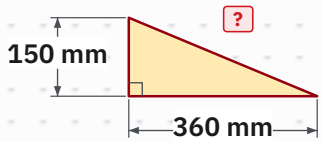
Find every red ?. Break it down where the drawing shows a dashed line.

Name _____

Date _____

1 Find the hypotenuse.

SHAPE CANVAS PRACTICE: FIND THE ?

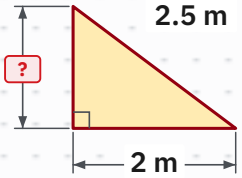


WORKING

Answer _____

2 Find the missing leg.

SHAPE CANVAS PRACTICE: FIND THE ?

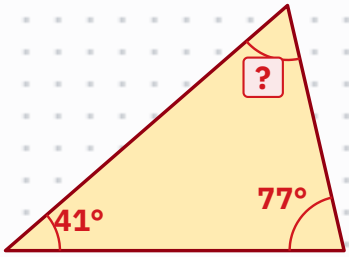


WORKING

Answer _____

3 Find the third angle.

SHAPE CANVAS PRACTICE: FIND THE ?

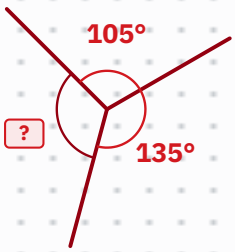


WORKING

Answer _____

4 Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

5 Find the ? on the bent strap.

SHAPE CANVAS PRACTICE: FIND THE ?

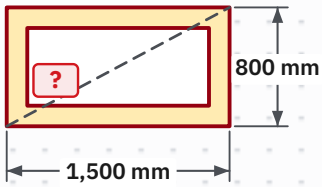


WORKING

Answer _____

6 Find the diagonal of the frame.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

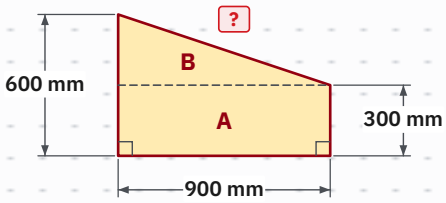
7 Change one value. A 300-400-500 triangle has its 300 mm leg made 600 mm long. The other leg stays 400 mm. Find the new hypotenuse to two decimal places. Did it double?

WORKING

Answer _____

8 Break it down. The dashed line splits the plate into rectangle A and right triangle B. Find the legs of triangle B, then the sloped edge to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

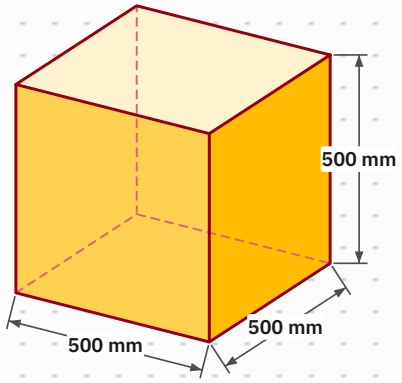


WORKING

Answer _____

9 Break it down. A cube has 500 mm edges. Find the diagonal of one face. Then use it and an edge as legs to find the longest rod that fits inside, corner to corner. Two decimal places.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

13 Trigonometry

EXERCISES

Practice naming sides, using sine, cosine and tangent, finding sides and angles, and solving slopes, pipe offsets, bevels and mitres. Your calculator must be in DEG mode.

EXERCISE 13A

Skills

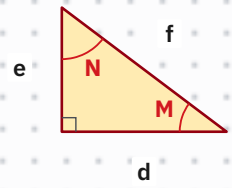
Sides, ratios, missing sides and missing angles.

Name _____

Date _____

1 Look at the triangle. Which side is opposite angle M? Which side is adjacent to angle M? Which side is the hypotenuse? Which side is opposite angle N?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

2 Find sin 25°, cos 25° and tan 25° to four decimal places.

WORKING

Answer _____

3 A right triangle has sides 8, 15 and 17. Angle A is opposite the 8. Write sin A, cos A and tan A to four decimal places.

WORKING

Answer _____

4 Hypotenuse 650 mm, angle 32°. Find the opposite side to two decimal places.

WORKING

Answer _____

5 Hypotenuse 1,250 mm, angle 48°. Find the adjacent side to two decimal places.

WORKING

Answer _____

6 Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.

WORKING

Answer _____

7 Opposite 520 mm, angle 38°. Find the hypotenuse to two decimal places.

WORKING

Answer _____

8 Opposite 300 mm, angle 61°. Find the adjacent side to two decimal places.

WORKING

Answer _____

9 Hypotenuse 42″, angle 15°. Find the opposite side to the nearest 1/16″.

WORKING

Answer _____

10 Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.

WORKING

Answer _____

11 Opposite 0.45 m, hypotenuse 1.2 m. Find the angle to two decimal places.

WORKING

Answer _____

12 Adjacent 700 mm, hypotenuse 950 mm. Find the angle to two decimal places.

WORKING

Answer _____

13 One acute angle of a right triangle is 27.4°. Find the other one.

WORKING

Answer _____

14 A slope rises 120 mm over a run of 6,000 mm. Find the percent grade and the angle to two decimal places.

WORKING

Answer _____

15 tan A = 0.25. Find A to two decimal places.

WORKING

Answer _____

EXERCISE 13B

Shop problems

Pipe offsets, slopes, bevels and mitres. Metric and imperial.

Name _____

Date _____

1 A line moves over 375 mm with 45° fittings. Find the travel and the run to two decimal places.

WORKING

Answer _____

2 A line moves over 9″ with 60° fittings. Find the travel and the run to the nearest 1/16″.

WORKING

Answer _____

3 A line moves over 12″ with 22.5° fittings. Find the travel to the nearest 1/16″.

WORKING

Answer _____

4 An offset made with 30° fittings has a travel of 640 mm. Find the set and the run to two decimal places.

WORKING

Answer _____

5 A line must move over 250 mm in a run of 433 mm. What fitting angle does it need, to two decimal places?

WORKING

Answer _____

6 A rolling offset rises 450 mm and rolls 600 mm. Find the true offset, then the travel with 45° fittings, to two decimal places.

WORKING

Answer _____

7 A ramp must rise 750 mm. The job sheet sets its angle at 4.8°. How long is the run, to the nearest millimetre?

WORKING

Answer _____

8 A drain line is laid at a 1% grade over an 18 m run. How far does it fall, in millimetres?

WORKING

Answer _____

9 A conveyor 8,500 mm long slopes up at 12°. Find the rise and the floor length to two decimal places.

WORKING

Answer _____

10 A 25 mm plate is bevelled at 30° through its thickness. How wide is the bevel at the top, to two decimal places? What groove angle do two such plates open?

WORKING

Answer _____

11 A frame has 10 equal sides. Find the corner angle, the cut angle to the edge and the angle away from a square cut.

WORKING

Answer _____

12 A truss top chord rises at 18° over a half span of 3,600 mm. Find the rise and the chord length to two decimal places.

WORKING

Answer _____

13 A 14 mm plate has a bevel 8 mm wide at the top, through the full thickness. Find the bevel angle to two decimal places.

WORKING

Answer _____

14 With the 45° travel constant 1.414, what travel do you get for a 375 mm set? How far is it from the calculator value in problem 1, to two decimal places?

WORKING

Answer _____

EXERCISE 13C

Shape Canvas

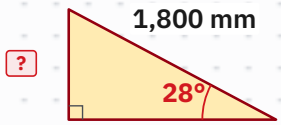
Every triangle is drawn at its true angle. Find every red ?

Name _____

Date _____

1 Find the height to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

2 Find the hypotenuse to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

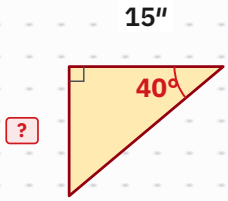


WORKING

Answer _____

3 Find the opposite side to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?

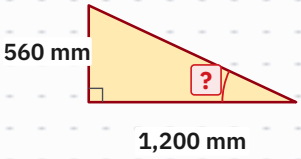


WORKING

Answer _____

4 Find the angle to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

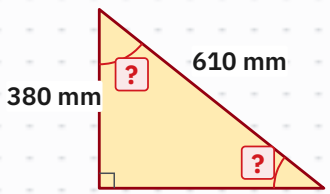


WORKING

Answer _____

5 Find both acute angles to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

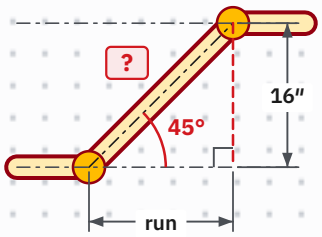


WORKING

Answer _____

6 A 45° offset. Find the travel to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?

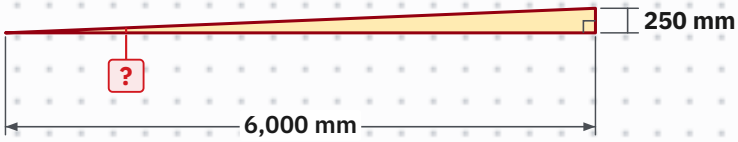


WORKING

Answer _____

7 A ramp rises 250 mm over a run of 6,000 mm. Find its angle to two decimal places. The drawing is to scale.

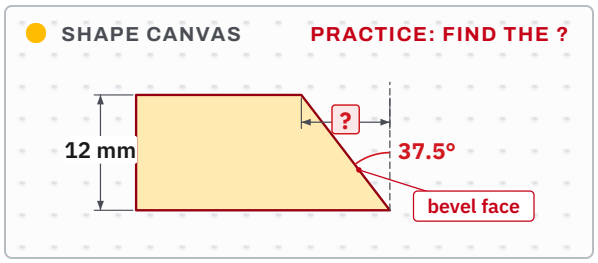
SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

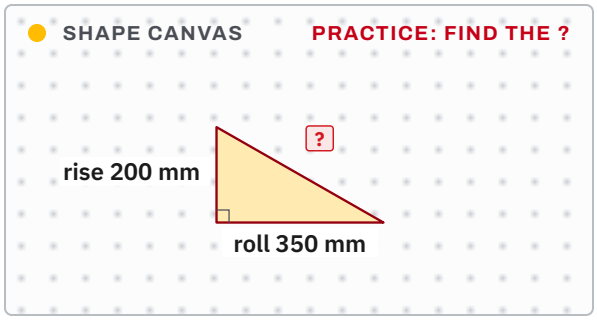
8 Find the bevel width to two decimal places.



WORKING

Answer _____

9 Break it down. A rolling offset rises 200 mm and rolls 350 mm. Find the true offset, then the travel with 60° fittings. Two decimal places.



WORKING

Answer _____

10 Change one value. A 45° offset with a 300 mm set has a travel of 424.26 mm. Double the set to 600 mm. What is the travel, to two decimal places? Did it double?

WORKING

Answer _____